Beyond few harmonics: the lower side of the logarithmic law of the resolution gap for windows with a weak ripple

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Paper F [1] proved the logarithmic law of the resolution gap with the sharp constant for periodic windows with at most twenty harmonics or with decay exponent $K \geq 0.003$ (which includes every window with $x \geq 0.003$): with $\ell = \ln(1/\sigma)$, the smallest gap obeys explicit two-sided bounds on $g_{\min}\sqrt{\ell}$ for $10^{-100000} \leq \sigma \leq 10^{-24}$, and its limit lies between $1 - \bar{\varepsilon}$ and 1, where $\bar{\varepsilon}$ bounds the remainder on the window class. Here we extend the lower side to windows with any number of harmonics whose ripple is weak in the sup norm or poor in low-frequency energy. Three tools do the work. A sharper treatment of one remainder of the curvature step turns an absolute error of size $e^{-\tau/2}/k'$ into a relative one. A split sandwich lets a low-frequency block enter through its energy rather than through its sup. Cell-based computer-assisted proofs, with an explicit bound on floating-point rounding in every acceptance test, certify the full lower bound, remainders included, and a mean-square bound of the curvature cross term closes the edge of very wide focus. The main theorem gives $g\sqrt{\ell} \geq (1 - \bar{\varepsilon})(\lambda/\Upsilon)(1 + 12.1/\ell + 2.5 \times 10^{-4})$ for every state with $\tau \geq 20$, for $\ell \in [55.26, 960.8]$, that is, for σ from 10^{-24} down to about 10^{-417}. In this range the coefficient one holds, up to the factor $1 - \bar{\varepsilon}$, for every window with $\eta \leq 1/\ell$, for a family of intermediate windows, and for seventeen thin-block windows certified one by one. A conjecture of our own (that $\eta e^{\mu/2}$ stays bounded) is false; we give explicit counterexamples, and they are covered by the theorem. We close with the road to an unconditional theorem, from the nearest restriction to the farthest. For the states with a narrow focus ($\tau < 20$) we prove that positivity alone makes the local brightness pay for fineness, that a ripple hidden at the centre of the focus is dark, and that the local route costs almost no remainder; with these bounds the envelope exceeds the target at every ℓ computed, but the cells that would turn this into a proof are not yet done. Further out lie the full bound beyond $\ell = 960.8$ and the limit $\sigma \to 0$, windows outside the covered families, the class constant $\bar{\varepsilon}$, and the upper side. We list exactly what is proved, what is computer-assisted, and what is only numerical. This version (v1.2) corrects two steps of the proof of the main theorem, the lower end point in the brightness G and the step that restores the sandwich factors, and closes both by re-running the certificates; the statement and the range of the theorem are unchanged. Version 1.1 closed three gaps in the covering of the computer-assisted proofs of v1.0 (a wedge of states inside each column of the main certificate, the extreme values of the energy frequency μ, and the smallest b of the reduced certificate), added a line-by-line covering argument with a random audit of 10^5 states, and corrected two statements of v1.0 about the narrow-focus states.

I. INTRODUCTION

The picture. An observer looks at a periodic pattern through a lens of finite resolution σ. The finer the pattern, the less of it survives the blur. Paper F [1] measured how much "information about the direction of time" a window keeps, through the quantity g. It showed that the smallest gap $g_{\min}$ falls like $1/\sqrt{\ell}$, $\ell = \ln(1/\sigma)$, with the coefficient one, for windows with at most twenty harmonics or with decay exponent $K \geq 0.003$ (which includes every window with $x \geq 0.003$). Precisely, $g_{\min}\sqrt{\ell}$ obeys explicit two-sided bounds for $10^{-100000} \leq \sigma \leq 10^{-24}$, its limit lies between $1-\bar{\varepsilon}$ and 1 on every class with remainder bound $\bar{\varepsilon}$, and it tends to one along an explicit comb. In the language of this paper, the coefficient one is *the exchange rate between fineness and blur*. Every factor e gained in fineness costs the same amount of brightness under the blur, and what remains is a fixed fraction $1/\sqrt{\ell}$.

What is new. Paper F needed either a bound on the number of harmonics ($N \leq 20$) or a lower bound on their damping ($K \geq 0.003$). A window with many harmonics can hide its ripple in two ways. It can spread its energy thinly over a wide band, so that the sup η of the ripple stays small. Or it can put a coherent block at low frequency, so that η is large but the energy is tiny. The second way is the reason why a conjecture we used as a working hypothesis is false (Sec. IX). That conjecture was $\eta e^{\mu/2} \leq C$, where μ measures the frequency of the ripple energy. We built explicit windows with $\log_{10}(\eta e^{\mu/2})$ up to 32.0 (App. B). The coefficient one survives on all of them; only the old tool (the η sandwich) failed.

How we prove it.

- Lemma R and Lemma A4* (Sec. III) give a lower bound of g for every state. All remainders are explicit, and the remainder a_4 of the curvature step (Lemma 23 of [1]) becomes relative to the ripple.

- Lemma S (Sec. IV) replaces the sup of a low-frequency block by its energy.

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- Cell-based computer-assisted proofs (Sec. VI) show that this lower bound exceeds the target.
- Lemma BE (Sec. VII) closes the small-b edge by measuring the curvature cross term in mean square instead of in sup.
- The boxes and Lemma BE cover every state of the theorem; this is shown line by line and checked on 10^5 random states, a check that also finds every deliberately removed box (Sec. VI).

Table I lists every result with its status. Throughout we keep the distinction of [1]: proof, computer-assisted proof, derivation, numerical, conjecture.

Picture. The lens sees a fine ripple only through its blurred brightness. This paper shows that the brightness that survives does not depend on how many harmonics make up the ripple, only on how much energy they carry at each frequency.

II. SETTING

We use the notation of [1] and recall only what is needed.

Windows and states. A periodic window is $p = \mathcal{N}_V \, \Gamma \, h/Z$. Here $\mathcal{N}_V$ is a Gaussian of variance $V = \sigma^2/b$ in v and $\Gamma = N(0, L^2)$ in w, and

$$h - 1 = e^{-w^2/2s^2} R(t), \qquad t = k'v, \tag{1}$$

$$R(t) = \sum_{n \geq 1} q0_n \cos(nt - \varphi_n). \tag{2}$$

The blurred coefficients are $q0_n = |a_n| e^{-n^2 K/2}$ with $K \geq x$, $x = (k'\sigma)^2$. We write

$$\eta = \sup |R|, \quad S_0 = \sum q0_n^2, \quad S_2 = \sum n^2 q0_n^2, \tag{3}$$

together with $\mu = xS_2/S_0$, $P = xS_2$, $\tau = x/b$, and $\bar{Q} = 1/4$ for the bound of (P3). A state is a choice of (b, L^2, s^2) subject to (P0) $L^2 \leq 1/\sigma$ and (P4) $\sigma^2 \leq s^2 \leq L^2/4$. The conditions (P0)–(P7) are those of Table 5 of [1].

The quantity g. As in Eq. (3) of [1], $g \geq (1 - |\varepsilon|)\,((a) + (a'))$, with

$$(a) = \frac{2A_w}{\sigma \, \mathrm{den}}, \qquad (a') = \frac{N_\phi}{\sigma \, \mathrm{den}}, \tag{4}$$

where $\mathrm{den} = \sigma^2(J_{vv} + J_{ww})$ and $|\varepsilon| \leq \bar{\varepsilon}$ is the window-class constant of (P6). The statement "$g\sqrt{\ell} \geq 1$" always means "up to the factor $1 - \bar{\varepsilon}$".

Regime. This paper works in $x \leq 0.003$ (the case $x \geq 0.003$ is Theorem 48 and Corollary 49 of [1]) and $x \geq 5 \times 10^{11}\sigma^2$, so that $\beta \leq 10^{-6}\tau$. The states have $\tau \geq 20$, i.e. the v-focus spans at least about 0.7 ripple periods (for $\tau \geq 80$ the remainders of Lemma 23 of [1] are already below 10^{-14}). The states with $b < 10^4\sigma^2$ (the small-b

TABLE I. Results of this paper and their status.

result	status
Lemma R (lower bound for every state)	proof (Sec. III)
Lemma A4* (a_4 relative)	proof
Lemma H (high part and θ_τ kernels)	proof (App. A)
τ cells, factor 0.99977 on $[20, 80]$	computer-assisted (App. A)
τ factors decrease on $\tau \geq 20$	proof (App. A)
Lemma S, Lemma S$'$, Corollary S	proof (Sec. IV)
x-free envelope (Lemma Z3$'$)	proof (Sec. V)
main reduced inequality, $\ell \leq 6907.8$, with the μ tails and $b \to 0$	computer-assisted (Sec. VI)
full Lemma R* bound, $\tau \geq 20$, $b \geq 10^4\sigma^2$, $\ell \leq 960.8$, with the μ tails and the wedge	computer-assisted (Sec. VI)
covering of the states by the boxes and Lemma BE	proof (line by line); random check of 10^5 states, numerical (Sec. VI)
Lemma BE (every state with $b < 10^4\sigma^2$)	proof (Sec. VII)
seventeen thin-block windows	computer-assisted (Sec. VI)
Theorem H1 (every state with $\tau \geq 20$)	computer-assisted, modulo (P6) (Sec. VIII)
envelope values (1.7116 at 10^{-24}, 1.0063 at 10^{-3000})	numerical
conjecture C ($\eta e^{\mu/2} \leq C$)	false (App. B)
local corrections (L1)–(L3), Lemma A*	proof (Sec. IX A)
local Z3$'$ with one universal constant	false; ratio 26.9 and 71.4 on explicit packets (Sec. IX A)
Lemma CM-1 (local brightness)	proof (Sec. IX A)
Lemma CN-1 and the joint bounds Ψ_2, Ψ_2^*	proof (Sec. IX A)
Lemma CN-2 (local F term)	proof (Sec. IX A)
Lemma R$_\theta$ (local route, explicit remainders)	proof (Sec. IX A)
sharp joint bound $\Psi \min(1, \sqrt{2\kappa})$	conjecture; survives the search (App. F)
local envelope with Ψ_2^* above T_0 ($\ell \leq 960.8$)	numerical (Sec. IX A)
states with $\tau < 20$	open (Sec. IX A)
full bound for $\ell > 960.8$, limit $\sigma \to 0$	open (Sec. IX B)

edge) are treated separately in Sec. VII; there $\tau \geq 5\times10^7$ automatically. The states with $\tau < 20$ are not covered (Sec. IX A).

Picture. b is how sharply the observer focuses along v, L how wide she looks along w, and s how wide the ripple is in w. τ counts how many ripple periods fit into the focus.

III. THE LOWER BOUND FOR EVERY STATE

Lemma 1 (Lemma R)**.** *[proof] For every state,* $(a) + (a') \geq \mathrm{Num_{lo}}/(\sigma\,\mathrm{den_{up}})$. *The bounds are built from Lemmas 19, 21, 22 and 23 of [1] (called BL0, BL1, BL2 and BL3 in its App. D) and four steps:*

- *(R1) the η sandwich for J_{per} and W_{per};*
- *(R2) the upper bound of* den*, from Lemmas 21 and 22 of [1] with (R1);*
- *(R3) a bound on the correction of A_w, using $\phi \leq (v'^2 + R^2)/2$ and $R^2 \leq 2w^2 + 4\chi^2$;*
- *(R4) the term $e\sigma^2\langle\theta_\tau F(1-\cos\psi)\rangle$ that Lemma 23 of [1] drops, kept as the main part of (a').*

The explicit form is in `ca_reduction.py`. At the hardest band points it agrees with the reduced formula of the earlier stage (`by_eval.py`, built from the lemmas of App. D of [1]) to 4.9×10^{-12} [numerical].

The remainder a_4. In Lemma 23 of [1] the F-term of the exact identity is split as

$$\sigma^2\big(G[F] - \mathrm{Re}\,A[Fe^{i\psi}]\big) = (1-e)\sigma^2\langle\theta_\tau F\rangle + e\sigma^2\langle\theta_\tau F(1-\cos\psi)\rangle + e\sigma^2\,\mathrm{Re}\langle(\theta_\tau-\Theta)Fe^{i\psi}\rangle. \quad (5)$$

Paper F bounds the last term by $\sup F$ times the norm of $\theta_\tau - \Theta$. This gives

$$a_4 = \frac{\sigma\tau\sqrt{x}\,N_1^2\nu_1}{\sqrt{2}(1-\bar{Q})}, \quad (6)$$

an absolute term of size $e^{-\tau/2}/k'$. It is harmless for the windows of [1], where the number of harmonics or their damping keeps N_1 bounded and the remainders are certified directly. For windows with many harmonics and $\sigma \leq 10^{-48}$ it makes the background part of the bound negative at $\tau = 20$.

Lemma 2 (Lemma A4*)**.** *[proof] With $\beta^2 = \tau y$ and $\bar{\chi}(w)$ the bound of $\sup_t|\chi|$ of Lemma 23 of [1],*

$$e\sigma^2\Big|\int \Gamma\,\mathrm{Re}\langle(\theta_\tau-\Theta)Fe^{i\psi}\rangle\Big| \leq a_4^* = e\,W_{\mathrm{per,up}}\Big[\frac{\beta^2\nu_2}{2} + \beta\nu_1\Big(\frac{\xi+\zeta}{\sqrt{2}} + \sigma^2\sqrt{\frac{s_{e2}}{\pi}}\,\frac{1}{L^2}\Big)\Big]. \quad (7)$$

Proof. Fix w, put $g = Fe^{i\psi}$, and let $\hat{g}_n$ be its Fourier coefficients. Pairing n and $-n$ gives

$$\mathrm{Re}\langle(\theta_\tau-\Theta)g\rangle = \sum_{n\geq1} e_n\,\mathrm{Re}\big[(\hat{g}_n + \hat{g}_{-n})(1-\cosh n\beta) + (\hat{g}_n - \hat{g}_{-n})\sinh n\beta\big]. \quad (8)$$

Since $F \geq 0$, $|\hat{g}_n| \leq \langle F\rangle$. With $\cosh u - 1 \leq (u^2/2)e^u$, the first sum is at most $(\beta^2\nu_2/2)\langle F\rangle$. In the second sum write $e^{i\chi} = 1 + (e^{i\chi}-1)$. The part with 1 is $-2\sin(w/\sqrt{2})\,\mathrm{Im}\hat{F}_n\sinh n\beta$. F is even in w, so its integral against the centred Γ vanishes exactly; this is the parity argument used in the proof of Lemma 23 of [1] for a_{10}. The details are in App. D 3. The rest is at most $\beta\nu_1\sup_t|\chi|\,\langle F\rangle$, because $\sinh u \leq ue^u$. Finally integrate against Γ; note $\sigma^2\int\Gamma\langle F\rangle = W_{\mathrm{per}}$. □

a_4^* is proportional to the ripple and moves to the ripple part of the bound. Relative to the coefficient $1-e$ of W it is of order $\tau\nu_2$ and $2\nu_1U\sqrt{x}$; both are τ factors. On the band windows forced to $\tau = 20$, the background part changes as follows [numerical]:

σ	$T_A\sqrt{\ell}$ with a_4	$T_A\sqrt{\ell}$ with a_4^*
10^{-48}	-2.6×10^{26}	2.7×10^{23}
10^{-300}	-8.8×10^{213}	1.2×10^{149}

The ripple part stays at $T_B\sqrt{\ell} \geq 1.054$. We call Lemma R with a_4 replaced by a_4^* *Lemma R*.*

Picture. The old bound counted the ripple's worst point at full weight even where the observer's focus averages it away. The new bound lets the average act first.

IV. SPLIT SANDWICHES

The η sandwich of [1] (Table 2 and App. A there) reads

$$\frac{\langle Y^2\rangle}{1+\eta} \leq \Big\langle\frac{Y^2}{h}\Big\rangle \leq \frac{\langle Y^2\rangle}{1-\eta} \quad (9)$$

for $Y = h_w$ and $Y = h_t$. It costs a factor $(1-\eta)/(1+\eta)$ in the bound. A coherent low-frequency block can make η large while carrying almost no energy, and then this factor is far too pessimistic.

Split $R = R_{\mathrm{lo}} + R_{\mathrm{hi}}$, where R_{lo} has frequencies in $[1, M]$ and R_{hi} frequencies $\geq m$. Let $Y = \phi(w)y(t)$ with $y = R$ or $y = R'$, and $h = 1 + \psi(w)R$ with $0 \leq \psi \leq 1$. Let $\langle\cdot\rangle$ be the period average integrated against a positive measure in w. Put

$$f_{\mathrm{lo}} = \frac{\|Y_{\mathrm{lo}}\|^2}{\|Y\|^2}, \quad \eta_{\mathrm{lo}} = \sup|R_{\mathrm{lo}}|, \quad \eta_{\mathrm{hi}} = \sup|R_{\mathrm{hi}}|, \quad (10)$$

$$\kappa = \min\Big(\eta_{\mathrm{lo}},\ \frac{\sup|y_{\mathrm{hi}}|\,\|R_{\mathrm{lo}}\|_t}{\|y_{\mathrm{hi}}\|_t}\Big), \quad (11)$$

$$E = \eta_{\rm lo} f_{\rm lo} + 2\eta_{\rm hi}\sqrt{f_{\rm lo} f_{\rm hi}} + (\eta_{\rm hi} + \kappa) f_{\rm hi}, \quad (12)$$

$$B = \big(\eta_{\rm lo}\sqrt{f_{\rm lo}} + \kappa\sqrt{f_{\rm hi}} + \eta_{\rm hi}\big)^2. \quad (13)$$

Lemma 3 (Lemma S)**.** *[proof] If $2M < m$ (a spectral gap), then*

$$\|Y\|^2 \max\Big(\frac{1}{1+\eta}, 1-E\Big) \le \Big\langle \frac{Y^2}{h} \Big\rangle \le \|Y\|^2 \min\Big(\frac{1}{1-\eta}, 1+E+\frac{B}{1-\eta}\Big). \quad (14)$$

Proof. Use $1/(1+r) = 1-r+r^2/(1+r)$ with $0 \le r^2/(1+r) \le r^2/(1-\eta)$, and expand $Y^2 r$. Two of the six products vanish because of the gap: $\langle Y_{\rm lo}^2 r_{\rm hi}\rangle$ and $\langle Y_{\rm lo} Y_{\rm hi} r_{\rm lo}\rangle$. The others are bounded by Cauchy–Schwarz. For the product $\langle Y_{\rm hi}^2 r_{\rm lo}\rangle$ use $\langle y_{\rm hi}^2 |R_{\rm lo}|\rangle_t \le \sup|y_{\rm hi}|\,\|y_{\rm hi}\|_t \|R_{\rm lo}\|_t$. This is where the energy $\|R_{\rm lo}\|_t^2 = S_{0,\rm lo}/2$ of the low block replaces its sup. For r^2 use Minkowski; the term $\|Y_{\rm hi} r_{\rm lo}\|$ is bounded with $\|y_{\rm hi} R_{\rm lo}\|_t \le \sup|y_{\rm hi}|\,\|R_{\rm lo}\|_t$ (or by $\eta_{\rm lo}$), so the same κ enters B as E. The line-by-line proof, with the norms and measures, is in App. D 2. □

Lemma 4 (Lemma S′, no gap)**.** *[proof] Without the gap, Lemma 3 holds with E replaced by*

$$E' = E + \eta_{\rm hi} f_{\rm lo} + 2\eta_{\rm lo}\sqrt{f_{\rm lo} f_{\rm hi}}. \quad (15)$$

Proof. The two products that used the gap are bounded by $\eta_{\rm hi}\|Y_{\rm lo}\|^2$ and $\eta_{\rm lo}\|Y_{\rm lo}\|\|Y_{\rm hi}\|$. The upper bound B used Minkowski only, so it is unchanged. □

Scaling. Let $\lambda = \min(\mathrm{low}_{S_0}, \mathrm{low}_{S_2})$ and $\Upsilon = \mathrm{up}_{S_2}$ be the lower and upper factors of the two channels. The part of den that carries S_0 is given its factor $1/(1-\bar\eta) = 2$ directly in the cells of Sec. VI. Write the lower bound of Lemma R* as $P_{\rm bg} + \mathrm{low}_{S_0} P_0 + \mathrm{low}_{S_2} P_2 - N$ and its den as at most $d_{\rm bg} + \Upsilon d_2$. Here $P_{\rm bg}$ collects the gains without a sandwich factor, $P_0, P_2 \ge 0$ the gains of the two channels, $N \ge 0$ the remainders, and $d_{\rm bg}, d_2 \ge 0$. All of them depend on the state only. Since $P_0, P_2, d_{\rm bg} \ge 0$ and $\Upsilon \ge 1$,

$$\mathrm{Num} - \frac{\lambda}{\Upsilon}\gamma\sigma\,\mathrm{den} \;\ge\; T(\lambda), \qquad T(\lambda) := P_{\rm bg} + \lambda(P_0+P_2) - N - \lambda\gamma\sigma(d_{\rm bg}+d_2). \quad (16)$$

T is affine in λ, so $T \ge 0$ on $[\lambda_*, 1]$ follows from $T(1) \ge 0$ and $T(\lambda_*) \ge 0$. Every window of Theorem 8 has $\eta \le \sum q 0_n \le \bar Q = 1/4$ by (P3), so its lower factors are at least $1/(1+\eta) \ge 4/5$ (the η sandwich, and the first entry of the maximum in Lemma 3), and its upper factor is at least one. We take $\lambda_* = 1/(1+\bar\eta) = 2/3$. $T(1) \ge 0$ is the certificate with the factors set to one. $T(\lambda_*) \ge 0$ is a second certificate, with the factors $(\lambda_*, \lambda_*, 1)$ and the target $\lambda_*\gamma$ (Sec. VI). Together they give $g \ge (1-\bar\varepsilon)(\lambda/\Upsilon)\gamma$ for every admissible window. The step cannot be shortened to "a certificate with the factors set to one gives $g \ge (\lambda/\Upsilon)\gamma$": that would need $-N \ge -\lambda N$, which fails for $\lambda < 1$. In general $T(\lambda) = \lambda T(1) + (1-\lambda)(P_{\rm bg} - N)$, and the second certificate is what controls $P_{\rm bg} - N$. Versions 1.0 and 1.1 stated the shortened step; see the version paragraph in Sec. VIII.

Corollary 5 (Corollary S)**.** *[proof] If the certified target is $T_{10} = 1 + 12.1/\ell + 2.5\times10^{-4}$ and $(\lambda/\Upsilon)T_{10} \ge 1$, then $g\sqrt{\ell} \ge 1$ (times $1-\bar\varepsilon$).*

Checks. On the nine windows that refute conjecture C, $(\lambda/\Upsilon)T_{10}$ lies between 1.0020 and 1.1074 [numerical]. These are exact sums over the window. A search for counterexamples to Lemmas 3 and 4 is described in App. B. It covered low and high frequencies, few and many harmonics, coherent and random phases, and used a hill climb. It found no violation. The same search does find violations of deliberately wrong versions in which $\eta_{\rm lo} f_{\rm lo}$, $\eta_{\rm hi} f_{\rm hi}$ or B is dropped. It finds none when the term $2\eta_{\rm hi}\sqrt{f_{\rm lo} f_{\rm hi}}$, the term $\kappa f_{\rm hi}$ or the gap-free cross term is dropped. Whether those three terms are needed is therefore not settled.

Picture. A low-frequency block is like a slow swell under the fine ripple. It can be tall, but the lens reads the ripple's detail through its energy, and a slow swell carries almost none of the detail.

V. THE X-FREE REDUCED INEQUALITY

The main reduced function. Put $A = \sigma^2/s^2$, $r = s^2/L^2 \in (0, 1/4]$, $g_s = \sqrt{r/(2+r)}$ and $G = P\,g_s$. The leading terms of Lemma R are

$$N = 2Ar + \frac{GA}{\mu(2+r)} + \frac{3\sigma^2}{8} + \frac{b\sigma^2}{8rA} + \frac{G\sigma^2}{16b} + \frac{G\sigma^2}{16A(2+r)}, \quad (17)$$

$$D = b + \frac{G}{2} + Ar + \frac{GA}{2\mu(2+r)}. \quad (18)$$

Here $g \ge \gamma$ holds if $N - \gamma\sigma D \ge 0$. N is of the form $\alpha A + \beta_1/A + \cdots$, so A can be eliminated exactly by the inequality of arithmetic and geometric means. The result is a function $\phi(\mu, r, b, G)$; $\phi \ge 0$ implies $g \ge \gamma$. For fixed (μ, r, b) it is concave in G, so the two end points $G = 0$ and $G = G_{\max}$ suffice (here the value at $G = 0$ is also the limit $G \to 0^+$; for the Lemma R* bound see App. E).

Lemma 6 (Lemma Z3′, x-free form)**.** *[proof] For $x \le 0.003$ and $\mu \ge 2$, $P \le \Pi^*(\mu)$, where*

$$\Pi^*(\mu) = \bar Q^2 \mu\,\frac{1+at_0}{(1+t_0)^2}, \quad (19)$$

with

$$t_0 = \frac{e}{2}(\mu - 0.003)\, e^{(\mu-0.003)/2}, \qquad a = c_s e^{-\mu/2}, \quad (20)$$

and $c_s = e^{0.0015}(1 + \sqrt{e} + 2/e) = 3.3896$ *(the value at* $x = 0.003$*). For every* $\mu > 0$*,* $P \le \mu \bar{Q}^2$*.*

The proof is the fractional knapsack argument of the earlier stages, with x removed at its largest value; it is written out in App. D 4. The monotonicity step needs $a \le 2$, i.e. $\mu \ge 1.055$; the cells use Π^* only for $\mu \ge 2$ and $\mu\bar{Q}^2$ below. Nothing is lost: the x-free envelope reproduces the envelope minimised over x. For example $g\sqrt{\ell} = 1.7116$ at $\sigma = 10^{-24}$, 1.3109 at 10^{-48}, 1.1508 at 10^{-96}, 1.0509 at 10^{-300} and 1.0063 at 10^{-3000} [numerical]. The envelope approaches one from above, roughly like $1 + 40/\ell$.

Picture. Of all ripples with the same energy frequency, the hardest one is the one that hides as much energy as the knapsack allows just below the blur. The envelope is the brightness that even this ripple must leave.

VI. COMPUTER-ASSISTED PROOFS

All cell proofs share one design. The parameters are cut into boxes in $(\log\mu, \log r, \log b)$; in the wedge of certificate (ii) the boxes are in $(\log\mu, \log r, \log\beta, \ell)$ with $\beta = b/\sigma^2$. On each box every term is bounded by its worst corner; each term is monotone in each variable. Terms with the same power of σ are combined at equal σ before the worst end of the ℓ cell is taken. Everything is computed in logarithms, and underflow may only drop non-negative parts. A box is accepted only when its margin exceeds a derived bound of the rounding error times the safety factor $S = 10$; otherwise it is bisected. The floating-point model, the bounds and the rule are stated in App. E. Tails in μ and b are separate boxes.

What the boxes cover. Each certificate claims an inequality on a set of states. The boxes, the tail boxes and the lemmas that take over parts of the set must cover it. The states of Theorem 8 are: $\ell \in [55.26, 960.8]$; $x \in [5 \times 10^{11}\sigma^2, 0.003]$; $\tau = x/b \ge 20$, so $0 < b \le b_{\max} = 1.5 \times 10^{-4}$; $\mu = xS_2/S_0 \in [x, \infty)$; $r = s^2/L^2 \in [\sigma^3, 1/4]$; $A \in [4\sigma^3, 1]$, minimised in closed form inside each box; and $G \in [0, G_{\max}]$, of which the two ends suffice by concavity; for the Lemma R* bound (certificates (ii) and (iii)) the lower end is the limit $G \to 0^+$ together with $G = 0$ itself (App. E). Every end that depends on σ refers to the state's own σ. A box of a column $[\ell_a, \ell_b]$ must therefore contain every state of the column, whatever its $\sigma \in [e^{-\ell_b}, e^{-\ell_a}]$. Write $\sigma_{\rm bot} = e^{-\ell_b}$ and $\sigma_{\rm top} = e^{-\ell_a}$.

Version 1.0 failed this in three places.

- In certificate (ii) the lower edge of b was $10^4\sigma_{\rm top}^2$, while Theorem 8 and Lemma 7 split the states at $10^4\sigma^2$. The wedge $10^4\sigma^2 \le b < 10^4\sigma_{\rm top}^2$ inside each column was covered by neither.
- The μ boxes of certificates (i) and (ii) stopped at e^{-30} and at $\mu_{\max}e^{60}$, $\mu_{\max} = \max(4\ell_b, 50)$.
- Certificate (i), whose claim is $b \in (0, 1]$, had no box below $b = e^{-2100-3\ell_b}$, and none below $\sigma_{\rm bot}^3$ for its two finite μ tails.

In none of them is there a sign that the inequality fails. Version 1.1 keeps every box of v1.0 and adds the boxes described below. The covering then reads, line by line:

- a state with $b < 10^4\sigma^2$ is covered by Lemma 7;
- a state with $10^4\sigma^2 \le b$ and $\beta < 10^4 e^{2(\ell_b-\ell_a)}$ lies in a wedge box (all $\mu > 0$);
- otherwise $b \ge 10^4 e^{2(\ell_b-\ell_a)}\sigma^2 \ge 10^4 e^{2(\ell_b-\ell_a)}\sigma_{\rm bot}^2 = 10^4\sigma_{\rm top}^2$, and the state lies in a box of v1.0 ($e^{-30} \le \mu \le \mu_{\max}e^{60}$) or in a μ tail box;
- $r \ge \sigma^3 \ge \sigma_{\rm bot}^3$ in every case.

A random check of the covering [numerical]. We drew 10^5 admissible states for (ii) and 5×10^4 for each of the two claims of (i), with extra weight near every edge (column ends, $b = 10^4\sigma^2$, $\tau = 20$, the ends of r, small and large μ, $b \to 0$). Each was looked up in the box tables of the code. None is missed. The same check finds the gaps in the v1.0 tables (30278 of 10^5 states for (ii)) and in each of eleven deliberately damaged tables: a column removed, the top of the wedge lowered by a factor e, a μ tail removed, the $b \to 0$ strip removed, and the like (`cq_cover.py`).

(i) The main reduced inequality. With $\gamma = T_D(\ell)/\sqrt{\ell}$ and $T_D = 1 + (2.1 + D)/\ell + 2.5 \times 10^{-4}$:

- for $D = 0$, the 301 columns $\ell \in [55.262, 3062.6]$ close, with 465900 audit points and no violation; the $D = 10$ columns cover $[55.26, 55.262)$, since $T_{10} \ge T_0$;
- for $D = 10$, all 686 columns $\ell \in [55.26, 6907.8]$ close ($\sigma = 10^{-24}$ to 10^{-3000}), with 1111902 audit points and no violation.

A column needs up to 7.8×10^7 boxes. Every accepted box has $\hat{\ell}_{\rm pos} - \hat{\ell}_{\rm neg} \ge S\delta$ (App. E); the smallest ratio $(\hat{\ell}_{\rm pos} - \hat{\ell}_{\rm neg})/(S\delta)$ over all accepted boxes is 1.00, as it must be, since a box is bisected until the rule holds. The tails $\mu \in (0, e^{-30}]$ and $[\mu_{\max}, \infty)$, with every $b \in (0, 1]$, and the strip of b below the v1.0 boxes for the other μ, close on all 301 + 686 columns (3.85×10^7 boxes). [proof, computer-assisted: `cd_cells.py` with the rule of `cl_cells.py`; tails `cq_core.py`] The certificate is not vacuous: at $\ell = 55.26$ the target 1.70 (below the envelope 1.711) closes, and 1.73 does not.

(ii) The full Lemma R bound.* Here all remainders of Lemma R* are kept:

- the corrections of A_w;
- the saturation of Ψ and Φ;

- both forms of the F remainder;
- a_4^*;
- a_1–a_{10} without a_4;
- the ν factors at $\tau = 20$.

The window enters only through generic bounds: $\sqrt{x}N_1 \le \bar{Q}\sqrt{2\ell+3.1}+10^{-6}$, $xN_2 \le \bar{Q}(2\ell+3.1)+10^{-4}$, and $\bar{\eta} = 1/2$ in the remainders. For $\tau \ge 20$ and $b \ge 10^4\sigma^2$, the 92 columns $\ell \in [55.26, 960.8]$ close with target T_{10} (σ down to about 10^{-417}), in three parts:

- the boxes of v1.0 in $(\log\mu, \log r, \log b)$, for $b \ge 10^4\sigma_{\rm top}^2$ and $\mu \in [e^{-30}, \mu_{\max}e^{60}]$ (2.09×10^9 boxes);
- the μ tails $(0, e^{-30}]$ and $[\mu_{\max}, \infty)$ for the same b (5.39×10^7 boxes, 15 CPU minutes);
- the wedge $10^4 \le \beta \le 10^4 e^{2(\ell_b-\ell_a)}$, every $\mu > 0$, in boxes of $(\log\mu, \log r, \log\beta, \ell)$ (4.44×10^8 boxes, 2.8 CPU hours; at most 1.67×10^7 boxes in a column).

Certificate (ii) is run twice: with the sandwich factors set to one and the target $\gamma = T_{10}/\sqrt{\ell}$, and with the factors $(2/3, 2/3, 1)$ and the target $\frac{2}{3}T_{10}/\sqrt{\ell}$ (the second certificate of the scaling step, Sec. IV). In the second run the target is written as $T_{D'}(\ell)/\sqrt{\ell}$ with $D' = (\frac{2}{3}T_{10}(\ell_a) - 1 - 2.5\times10^{-4})\ell_a - 2.1$ per column; then $T_{D'}(\ell) \ge \frac{2}{3}T_{10}(\ell)$ for every $\ell \ge \ell_a$. All 92 columns of the Lemma R* boxes, the 92 wedge columns and the 92 columns of the μ tails close in the second run as well, with the limit $G \to 0^+$ in every box (2.57×10^9 cells; smallest normalised margin 2.4×10^{-10}; a random check at intermediate G on 2.85×10^8 values finds no negative one). The cell counts differ slightly from those of the first run, because the acceptance differs. With the factors 2/3 but the target not lowered, the first column does not close, so the second run is not vacuous. [proof, computer-assisted: `cl_link.py` and `cq_core.py` with FAC set; wrapper `da_scaling.py`]

A wedge box all of whose states have $b \ge 10^4\sigma_{\rm top}^2$ belongs to the v1.0 boxes and is skipped (1.66×10^7 boxes). With β as a box variable but ℓ fixed per column, the first column does not close: in the wedge the main balance and the largest cost both grow like σ and are measured at opposite ends of the column. Subdividing ℓ removes this loss (App. E). Every term is moved by $S\eta_t + 1000Su$ in the safe direction before the terms are netted (App. E), so that the cancellation between equal powers of σ is covered. The states with $b < 10^4\sigma^2$ are covered by Lemma 7 (Sec. VII); the boundary $\beta = 10^4$ is a face of the wedge boxes. [proof, computer-assisted: `cl_link.py`; tails and wedge `cq_core.py`] Pointwise, the bound agrees with the main reduced function to about 2×10^{-4} near the envelope. Its soundness against a decimal evaluation of Lemma R* was checked at random states [numerical].

(iii) Seventeen thin-block windows. For the intermediate windows not covered by Corollary 5, the Lemma S factors are put inside the boxes. For each window, μ, x and P are fixed at the window's values, and $G \le Pg_s(r)$ is used exactly. All 17 windows close for the target $g\sqrt{\ell} \ge 1$, with the rule of (ii). Each window has a single σ, and its boxes cover $r \in [\sigma^3, 1/4]$ and $b \in [10^4\sigma^2, x/20]$ exactly. [proof, computer-assisted: `cf_windows.py` with `cl_link.py`] For one of them we checked that raising the target to 1.2 times the window's upper value makes the proof fail, as it must.

Picture. The computer does not search for the best state. It checks that no state, in any box, can do worse than the target, and every box keeps a cushion ten times larger than the largest possible rounding error. A second check asks the converse question: is every state the theorem talks about in some box? Version 1.1 answers it by construction and by 10^5 random states.

VII. THE SMALL-b EDGE

The certificates of Sec. VI cover $b \ge 10^4\sigma^2$. Below this edge the cells cannot work with the generic bound γ_P of the ripple derivative: the term $-\sigma^2\gamma_P\Phi$ of the curvature step is not tied to the size of the ripple. We close the edge by measuring the cross term of Lemma 23 of [1] (BL3) in mean square instead of in sup. The coefficient need not be one here; the bound is of order $1/\sigma$.

Lemma 7 (Lemma BE)**.** *[proof] Let $\sigma \in [10^{-3000}, 10^{-24}]$, $x \in [5\times10^{11}\sigma^2, 0.003]$, (P0), (P3), (P4), and $q0_n \le 2e^{-n^2x/2}$. Then every state with $0 < b \le 10^4\sigma^2$ satisfies*

$$(a) + (a') \;\ge\; \frac{\kappa}{\sigma}, \qquad \kappa = 1.1\times10^{-5}. \tag{21}$$

Hence $g\sqrt{\ell} \ge (1-\bar{\varepsilon})\,\kappa\sqrt{\ell}/\sigma \ge (1-\bar{\varepsilon})\,8\times10^{19}$. Condition (P2) is not used, and there is no restriction on x inside the regime. The bound on $q0_n$ holds for every window of [1]: a positive profile has $|a_n| \le 2$, and $q0_n = |a_n|e^{-n^2K/2}$ with $K \ge x$ (Sec. II).

The proof has four steps. Write $E = e^{-w^2/2s^2}$, so that $h = 1 + ER$.

(BE-a) The ripple derivative. Let $G = \partial_v(\tilde{s} + s_w)$ as in App. A of [1]. A direct computation, using $h - ER = 1$, gives

$$\sigma^2 G = E\Big[\frac{xR''}{h} - \frac{xER'^2}{h^2} - \sigma\sqrt{x}\,\frac{w}{s^2}\,\frac{R'}{h^2}\Big]. \tag{22}$$

Hence $|\sigma^2 G| \le E\,[xN_2(1-\bar{Q}) + xEN_1^2 + \sigma\sqrt{x}\,(|w|/s^2)N_1]/(1-\bar{Q})^2$. Its supremum over w is below the γ_P of [1] (Table 2 there). The envelope E shows that $\sigma^2 G$ lives where the ripple lives.

(BE-1) The cross term in mean square. By the parity identity $\langle hG\rangle = -\langle F\rangle$ (proof of Lemma 23 of [1]), for

every $c>0$ and every w,

$$\langle h\sigma^2 G\cos\psi\rangle \ge -\sigma^2\langle F\rangle - \frac{\langle h(\sigma^2 G)^2\rangle}{4c} - 2c\,\langle h(1-\cos\psi)\rangle. \quad (23)$$

Here we used $(1-\cos\psi)^2\le 2(1-\cos\psi)$. In Lemma 23 of [1] this replaces only the step $\ge -\sigma^2\langle F\rangle - \gamma_P\langle h(1-\cos\psi)\rangle$. The remainders a_5, a_6 keep γ_P; they carry ν factors.

(BE-2) The mean square. Put $M_G=\int\Gamma\langle h(\sigma^2G)^2\rangle$. From (BE-a), $\langle R''^2\rangle = S_4/2$, $\langle R'^2\rangle = S_2/2$, $\langle R'^4\rangle \le N_1^2 S_2/2$, $\int\Gamma E^2 = g_s$ and $\int\Gamma E^2 w^2/s^4 \le g_s/(2s^2)$,

$$M_G \le \frac{3g_s}{2}\Big[\frac{x^2S_4}{1-\bar Q}+\frac{xS_2\,(U_1^2+\frac12)}{(1-\bar Q)^3}\Big], \quad (24)$$

with $U_1\ge\sqrt{x}N_1$ the window constant of Sec. VI. For any $\Lambda>0$, $x^2S_4\le \Lambda\,xS_2+T_\Lambda$, where $T_\Lambda\le 4\sum_{xn^2>\Lambda}(xn^2)^2e^{-xn^2}$.

(BE-3) Choice and comparison. Take $c=(b+\sigma^2J_{ww})/(16\sigma^2)$ and $\Lambda=10^{-3}/\sigma^2$. With $\Phi\le 4\Psi$ and $\Phi\le 2$, the Φ part costs at most $b\Psi/2+\sigma^2J_{ww}/4$. The cross term enters Lemma 23 of [1] with a factor $e(1-b)\sigma^2$, so the M_G part is $e\sigma^2M_G/(4c)=4e\sigma^4M_G/(b+\sigma^2J_{ww})$. With $xS_2g_s\le 2(1+\bar Q)W_{\rm per}$ it is at most

$$\rho_W(1-e)W_{\rm per}+\frac{6\sigma^2g_sT_\Lambda}{(1-\bar Q)J_{ww}}, \quad (25)$$

where

$$\rho_W=\frac{48(1+\bar Q)K_1\sigma^2}{(1-b)^2}\le 0.0801, \quad (26)$$

and $K_1=\Lambda/(1-\bar Q)+(U_1^2+\frac12)/(1-\bar Q)^3$. Here $y=(1-b)^2\sigma^2/(2b)\ge 4.99\times10^{-5}$, so $1-e\ge y/(2+y)\ge 2.49\times10^{-5}$. With $2A_w\ge\sigma^2J_{ww}$ ($\phi\le2$) and $\sigma^2J_{vv}=b+W$ (Lemma 22 of [1]), this gives

$$2A_w+N_\phi \ge \frac78\sigma^2J_{ww}+0.45(1-e)W + \frac14 b\Psi - \text{rem}. \quad (27)$$

Every remainder (the a_i of Lemma 23 of [1] with a_4^*, the Φ-proportional terms, the ν terms of the W sandwich, and T_Λ) is below 10^{-10^7} times its comparison quantity. The reason is $\tau=x/b\ge 5\times10^7$. For $L^2\ge1$, $\Psi\ge\min(\Phi,\frac12)\ge 0.2211$. For $L^2<1$, $\sigma^2J_{ww}>0.6\sigma^2\ge 6\times10^{-5}b$. In both cases $2A_w+N_\phi\ge 1.14\times10^{-5}\,\text{den}$. Each step is written out in App. D 5. □

The constants, the formula of (BE-a) against finite differences, the bound of (BE-2) on random windows and every remainder are checked in `ch_bedge.py` [numerical].

Picture. When b is tiny the observer's v-focus is wide. Then the slope information W of the ripple enters almost undiminished. The cross term that blocked the edge was measured at the ripple's worst point. Measured in mean square, it is a few per cent of W.

VIII. MAIN THEOREM

Theorem 8 (Theorem H1)**.** *[proof, computer-assisted: Secs. VI and VII] Let a periodic window satisfy (P0), (P2)–(P4), (P6), $x\le 0.003$ and $x\ge 5\times10^{11}\sigma^2$, with $\ell\in[55.26,960.8]$. Then every state with $\tau=x/b\ge20$ satisfies*

$$g\sqrt{\ell}\ \ge\ (1-\bar\varepsilon)\,\frac{\lambda}{\Upsilon}\Big(1+\frac{12.1}{\ell}+2.5\times10^{-4}\Big). \quad (28)$$

Every state with $b<10^4\sigma^2$ satisfies the much stronger bound $g\sqrt{\ell}\ge(1-\bar\varepsilon)\,8\times10^{19}$ (Lemma 7). Here λ/Υ is the sandwich ratio of Sec. IV: $(1-\eta)/(1+\eta)$ for the η sandwich, or the Lemma S/S' factors. In particular, over the states with $\tau\ge20$, $g_{\min}\sqrt{\ell}\ge 1-\bar\varepsilon$ holds for:

(i) every window with $\eta\le 1/\ell$;

(ii) every window with $(\lambda/\Upsilon)T_{10}\ge1$, which includes the nine windows that refute conjecture C;

(iii) the seventeen thin-block windows of Sec. VI.

In the observer's words, the condition $\tau\ge20$ excludes only the states whose v-focus is narrower than about 0.7 ripple periods (precisely, $k'\sqrt{V}<\sqrt{20}$). These states are the subject of Sec. IX.

Proof. Split the states with $\tau\ge20$ at $b=10^4\sigma^2$.

States with $b\ge10^4\sigma^2$. Lemma 1 with Lemma 2 gives $g\ge(1-\bar\varepsilon)\,\text{Num}/(\sigma\,\text{den})$. Certificate (ii) of Sec. VI gives $T(1)\ge0$ and $T(2/3)\ge0$ in the notation of the scaling step, for $b\in[10^4\sigma^2, 0.003/20]$. Every state with $\tau\ge20$ has $b=x/\tau\le 0.003/20$. The set of these states is covered by the boxes of certificate (ii) (v1.0 boxes, μ tails and wedge) and by Lemma 7, line by line as in the covering paragraph of Sec. VI. The scaling argument of Sec. IV then gives $\text{Num}-(\lambda/\Upsilon)\gamma\sigma\,\text{den}\ge0$ for every admissible pair of factors.

States with $b<10^4\sigma^2$. Lemma 7 gives $(a)+(a')\ge 1.1\times10^{-5}/\sigma$, which is above γ by many orders of magnitude. The sandwich ratio is at most one, so the bound of the theorem follows.

For (i), $(1-1/\ell)(1+12.1/\ell+2.5\times10^{-4})/(1+1/\ell)\ge1$ on the whole range. For (ii) see Corollary 5. For (iii) see certificate (iii). □

Relation to [1]. Theorem 8 does not need a bound on the number of harmonics. Paper F covers windows with at most twenty harmonics or with damping exponent $K\ge0.003$ (Theorem 12 and Corollary 13 of [1], with Theorems 45 and 48 for the two classes), for $10^{-100000}\le\sigma\le10^{-24}$ and in the limit, with both sides of the law. Theorem 8 covers the lower side for windows with a weak or energy-poor ripple, for σ from 10^{-24} down to about 10^{-417} and for every state with $\tau\ge20$. The main reduced inequality alone (certificate (i)) holds up to $\ell=6907.8$.

Picture. However the ripple is built, many harmonics or few, the observer keeps at least the fraction $1/\sqrt{\ell}$,

as long as the ripple's sup or its low-frequency energy is small and her focus spans at least about 0.7 ripple periods.

Version 1.2. Two steps of the proof of Theorem H1 are corrected; the statement and the range of the theorem are unchanged. First, the rule that the two end points in G suffice was applied in v1.0 and v1.1 with $G = 0$ as the lower end. For the Lemma R* bound the lower end is the limit $G \to 0^+$, because one remainder bound adds a term without G on the branch $G > 0$ (App. E). Every certificate of Theorem H1 has been re-run with this limit; every box passes and every cell count is unchanged. Second, the scaling step that restores the sandwich factors (Sec. IV) used $-N \geq -\lambda N$ for the remainders, which fails for $\lambda < 1$. The corrected step needs one more certificate, the same certificate (ii) with the lower factors $2/3$ and the target $\frac{2}{3}\gamma$. It closes on all $92+92+92$ columns (Sec. VI). In v1.1 the missing condition had been checked only at random states (8×10^5 states, no failure). Both points were found by the author's own audit while working on the states with a narrow focus.

Version 1.1. Version 1.0 had three gaps in the covering of the computer-assisted proofs: the wedge $10^4\sigma^2 \leq b < 10^4 e^{-2\ell_a}$ inside each column of certificate (ii), the μ ranges below e^{-30} and above $\mu_{\max} e^{60}$ in certificates (i) and (ii), and the smallest b in certificate (i). Theorem H1 was therefore not proved in v1.0 for the states of the first two gaps. Version 1.1 closes all three: the wedge by boxes in (μ, r, β, ℓ) with $\beta = b/\sigma^2$ and ℓ itself subdivided, the μ tails by boxes reaching 0 and ∞, and the small b of certificate (i) by a strip down to $b = 0$ (Sec. VI, App. E). Every box of v1.0 is kept, all new boxes close, and a random check of 10^5 states finds the covering complete. The statement and the range of Theorem H1 are unchanged. Version 1.1 also corrects two statements of v1.0 about the states with a narrow focus (Sec. IX A): the conjectured local Z3′ with one universal constant is false, and the local form of Lemma R does not keep the algebraic form of Lemma R in the term $(1-e)W$. It adds the results on these states obtained since (Lemmas CM-1, CN-1, CN-2 and R_θ, and the envelope with the proved brightness bound), and proves the monotonicity of the τ factors on the whole range $\tau \geq 20$ (App. A).

IX. THE ROAD TO AN UNCONDITIONAL THEOREM

Theorem 8 has five restrictions. We list them from the nearest to the farthest, each with what is known and its status. The small-b edge is no longer among them: Lemma 7 closes it.

A. States with a narrow focus ($\tau < 20$)

Status. The states with $\tau < 20$ are *not* covered by Theorem 8, and they are not closed in this version. This subsection records what is proved about them, what is only numerical, and two corrections to v1.0.

What changes. When τ is small the observer sees the ripple only near the focus position t_0, through the wrapped normal θ_τ (Lemma 19 of [1]). Lemma R replaces these local averages by period averages and pays with factors $\nu_k \sim e^{-\tau/2}$. This route has a ceiling: the τ factor of the ripple part, $\sqrt{(1-\epsilon_1)(1-\nu_0)/(1+z)}\,(1-z)/(1+\nu_0)$, falls below T_{10}/env at $\tau \approx 5.4$ ($\ell = 55.26$), 10.1 ($\ell = 690.8$) and 14.1 ($\ell = 6907.8$) [derivation]. Below the ceiling, neighbouring high harmonics beat: their difference frequency is not averaged out by a narrow focus. For two neighbours $m = 50, 51$ the focused slope energy $\langle \theta_\tau R'^2 \rangle / \langle R'^2 \rangle$ ranges over $[0.005, 1.995]$ at $\tau = 0.01$, although $m^2\tau = 25$ [numerical]. So below the ceiling the local averages must be kept. We write $E_0 = \langle \theta_\tau R^2 \rangle_t$, $E_1 = \langle \theta_\tau R'^2 \rangle_t$, $\mu_\theta = xE_1/E_0$, $P_\theta = 2xE_1$ and

$$\kappa = \langle (s^2/\tau)\, \theta_\tau R'^2 \rangle_t / E_1, \tag{29}$$

with $s = t - t_0$ unfolded to the line. These are the local energy, slope energy, energy frequency, brightness and spread of the ripple at the focus.

Two corrections to v1.0. Version 1.0 conjectured a local Z3′ with one universal constant, $P_\theta \leq C\,\Pi^*(\mu_\theta)$, and stated that the largest ratio found numerically was 2.13. Both statements were wrong. A coherent packet of high harmonics placed at a node of the fundamental (it satisfies (P2), (P3) and positivity) gives $P_\theta/\Pi^*(\mu_\theta) = 26.9$ at $x = 0.003$ and 71.4 at $x = 10^{-6}$, and the ratio grows with μ_θ [numerical] (`cm_counter.py`). The value 2.13 came from a search whose packets had small raw coefficients. Version 1.0 also stated that the ripple part of the local form of Lemma R keeps the algebraic form of Lemma R. This fails for the term $(1-e)W$: in the local form it carries the factor κ (Lemma 11). So the narrow-focus states do not reduce to one local inequality. Theorem 8 is not affected: it concerns $\tau \geq 20$, where Lemma A4* bounds the remainder $\Theta - \theta_\tau$ and $\kappa \to 1$.

Local brightness [proof].

Lemma 9 (Lemma CM-1, local brightness)**.** *[proof] For every window of the class and every focus, with $E_0' = \langle \theta_{2\tau} R^2 \rangle_t \leq \bar{Q}^2$,*

$$xE_1 \leq \Big(1 + \frac{b}{2}\Big) E_0 \ln \frac{2B_0}{E_0} + \frac{\sqrt{2}}{e}\, b\, E_0', \qquad B_0 = \big(1 + \sqrt{E_0} + (E_0/2)^{1/(2+b)}\big)^2. \tag{30}$$

The proof uses only $f \geq 0$: analytic continuation of the blurred profile, Paley–Wiener for the Gaussian focus, and a one-parameter dual of the Chernoff bound for the second moment (`cm_local_bound.py`, `cm_checks.py`). Let $\Psi(\mu_\theta, b)$ be the largest P_θ compatible with the lemma; $\Psi \approx 4B_0\mu_\theta e^{-\mu_\theta}$ as $b \to 0$. So the correct form of the local Z3′ is $P_\theta \leq c\,\mu_\theta e^{-\mu_\theta}$, and it is proved: positivity alone makes local brightness pay for fineness.

Two foci [proof].

Lemma 10 (Lemma CN-1, two foci)**.** *[proof] For $\tau' \le \tau$ and $\delta^2 > \kappa\tau$,*

$$E_1 \le \sqrt{\tau'/\tau}\, e^{\delta^2/2\tau'}\, \frac{E_1^{(\tau')}}{1-\kappa\tau/\delta^2}, \tag{31}$$

$$E_0^{(\tau')} \le \sqrt{\tau/\tau'}\, E_0, \tag{32}$$

where the superscript (τ') denotes the same quantity with the focus $\theta_{\tau'}$.

Proof. By Chebyshev, the part of E_1 with $|s| > \delta$ is at most $(\tau/\delta^2)\kappa E_1$. For $|s| \le \delta$, $\theta_\tau(s) \le \sqrt{\tau'/\tau}\, e^{\delta^2/2\tau'}\theta_{\tau'}(s)$. The second inequality is $\theta_{\tau'} \le \sqrt{\tau/\tau'}\,\theta_\tau$ pointwise. □

With $\delta^2 = K\kappa\tau$ and $\tau' = L\delta^2$ ($K > 1$, $L > 0$, $LK\kappa \le 1$), Lemma 9 at the narrower focus gives the *joint bound* $P_\theta \le \Psi_2(\mu_\theta, b, \kappa) \le \Psi(\mu_\theta, b)$ [proof]. It decays like $\sqrt{\kappa}$ for $\kappa \le 10^{-3}$, with a constant 25–50 relative to $\Psi\sqrt{\kappa}$, and equals Ψ for $\kappa > 1/2$. Optimising (K, L) continuously instead of on a grid gives a smaller bound Ψ_2^*, still proved. A numerical search for the sharp joint bound found $P_\theta \le \min(2\mu_\theta\bar{Q}^2, \Psi\min(1, \sqrt{2\kappa}))$ on every window tried, with a true constant near 0.14 in the $\sqrt{\kappa}$ law [numerical] (App. F); this sharper form is not proved.

The local F term [proof].

Lemma 11 (Lemma CN-2, local F term)**.** *[proof] Let* $\mathrm{se2} = s^2/(2+r)$, $g_s = \int \Gamma e^{-w^2/s^2}$, $W_\theta = x g_s E_1$, $y = (1-b)^2\sigma^2/(2b)$, *and let ξ_e, $\bar{\chi}_e$ be the even and the remaining parts of the phase correction of App. D of [1]. Then the F term of the local identity satisfies*

$$\sigma^2\big(G[F] - \mathrm{Re}\, A[Fe^{i\psi}]\big) \ge \frac{W_\theta}{1+\eta}\Big[\big(1-e^{-\mathrm{se2}/4}\big) - \bar{\chi}_e + e^{-\mathrm{se2}/4}\Big(\frac{y}{2}\,\kappa - \xi_e\sqrt{y\kappa} - 0.51\,y^2\frac{E_1^{(2\tau)}}{E_1}\Big)\Big]. \tag{33}$$

Proof sketch. In the exact identity of App. D of [1], $A[g] = \mathbb{E}_\rho[e^{i\beta s/\tau} g]$ is a tilted Gaussian average on the line. The Gaussian average in w gives the factor $e^{-\mathrm{se2}/4}$. The even part of the phase correction is bounded with $|\sin(\beta s/\tau)| \le \beta|s|/\tau$ and Cauchy–Schwarz, which gives $\xi_e\sqrt{y\kappa}$. The chirp uses $1 - \cos z \ge z^2/2 - z^4/24$ and $(s^4/\tau^2)\theta_\tau \le 12.25\,\theta_{2\tau}$. □

The second bracket is the local form of $(1-e)W$: the factor $(1-e) \approx y/2$ carries κ. For $\tau \ge 20$ the focus covers the period, $\kappa \to 1$, and the bound of Lemma A4* is recovered.

The local route as a lemma with explicit remainders [proof]. Lemma BL3 of [1] rests on an exact identity, (BLid), for $\sigma^2 \int P s_v^2 \phi\, dv$. Reading it with the local averages kept gives the local form of Lemma R.

Lemma 12 (Lemma R_θ, the local route)**.** *[proof] For every periodic window state, $(a) + (a') \ge \mathrm{Num}_\theta/(\sigma\,\mathrm{den}_\theta)$. The main parts are those of Lemma R with (S_0, S_2) replaced by $(2E_0, 2E_1)$ and no factor ν_k. The F term is that of Lemma 11, so the term $(1-e)W$ carries κ. The remainders are seventeen explicit terms N_1–N_{17} in the numerator and D_1–D_3 in the denominator (`co_remainders.py`):*

- *the local normalisation $Z = 1 + k_E R_\theta$, $|R_\theta| \le \eta$, on the b terms;*
- *the focus term of the w information (L1), at most $2\bar{Q}\sqrt{r/(1+r)}/L^2$;*
- *the curvature term of W (L2), at most $(2/e)\bar{Q}\,b = 0.184\,b$;*
- *the corrections of A_w, with the ring-weighted average $\langle (s^2/\tau)\theta_\tau \cdot \rangle$;*
- *the first- and second-order chirp remainders of Lemma 11;*
- *the local failure of the parity identity $\langle hG\rangle = -\langle F\rangle$, a term $\sigma^3\sqrt{xE_1}$;*
- *the local forms of a_3, a_6, a_8, a_{10}, in which ν_1 becomes $\sqrt{(1+y)/\tau}$ and N_1 becomes $\sqrt{E_1}$ (Cauchy–Schwarz on the line), with the envelope factors of γ_P kept.*

The remainders a_1, a_2, a_5, a_9 and the factors ϵ_1, $1 \pm \nu_0$, z, ω_2 of Lemma R have no local counterpart.

Each new inequality was first attacked numerically, with deliberately wrong versions that the search broke [numerical]. The search covered packets at the node, crest and trough of the fundamental, flat and Gaussian envelopes, two packets, combs, 150 random bands, foci on and off the packet, and τ from 10^{-4} to 20; seven deliberately wrong versions were all broken. The whole lemma was then compared with direct two-dimensional quadrature on 46 states with τ from 0.3 to 20. It holds on all of them, and where Lemma R is negative it keeps 27–94 % of the exact value [numerical] (`co_rtheta.py`).

The remainders cost almost nothing [numerical]. Let D be the room left after the remainders of Lemma 12 are paid, in the target $T_D = 1 + (2.1 + D)/\ell + 2.5 \times 10^{-4}$. At the minima of the local envelope and over every admissible x, the sum of the remainders relative to the numerator is $D \le 1.2 \times 10^{-5}$ at $\ell = 55.26$ and far smaller at larger ℓ (`co_remainders.py`). So the target that is left after the remainders is T_0 itself.

The envelope with the proved brightness bound [numerical]. With $P_\theta \le \min(2\mu_\theta\bar{Q}^2, \Psi_2^*(\mu_\theta, b, \kappa))$, the κ-weighted W term and the corrections (L1)–(L3) at their worst, we minimised the local envelope over μ_θ, b, r, $\kappa \ge 10^{-30}$ and the width A by a multi-start Nelder–Mead search (`co_envmin.py`). It exceeds T_0 at every ℓ of the table: 1.1349 vs 1.0383 ($\ell = 55.26$), 1.0729 vs 1.0193 (110.5), 1.0394 vs 1.0098 (221), 1.0208 vs 1.0049 (455), 1.0144 vs 1.0033 (690.8), and 1.0107 vs 1.0024 (960.8). With the grid of (K, L) the values are 1.1343, 1.0723, 1.0365, 1.0150, 1.0071 and 1.0026. The nine windows that refute Conjecture C need $\Lambda_S T \ge 1$, i.e. $T \ge 1.022$, 1.014, 1.008

at $\sigma = 10^{-48}, 10^{-96}, 10^{-300}$; the same envelope passes there too. A uniform room $10/\ell$ for Corollary S, as in Theorem 8, is not available below $\ell \approx 960$ on this route. These are numerical minima, not certificates: the states with $\tau < 20$ remain open until the envelope is certified by cells.

Two corrections to our own first computations. A first computation of this envelope took $\kappa \geq 10^{-4}$ and concluded that a remainder budget $D \leq 2$ would make a joint bound unnecessary. That was wrong: as $\kappa \to 0$ the W term vanishes and the envelope without a joint bound falls to 0.95 ($\kappa \geq 10^{-12}$) and to the W-free value 0.81, below T_0; a κ-dependent brightness bound is needed whatever the budget, and Ψ_2 is one. A first grid minimisation of the envelope with Ψ_2 also missed the minimum at $\ell = 55.26$ (1.1845 instead of 1.1343); the values above come from the multi-start search, whose best four starts agree to 10^{-3}.

What remains for $\tau < 20$. The cells of the local route, in the five variables $(\mu_\theta, r, \beta, \kappa, \ell)$, with ℓ subdivided as in the wedge of certificate (ii). A first trial with ℓ fixed per column did not close, and only through the width in ℓ [numerical]. Also needed: the monotonicity of Ψ_2 inside a box, used at the corners and so far checked only numerically, and an independent audit of the cell code.

Lemma A [proof].* In Lemma 23 of [1], every remainder of the form $\nu \sup_t |g|$ may use $\nu\langle|g|\rangle$ instead. The kernels $\Theta - 1$, $\theta_\tau - \Theta$ and $\partial_t \Theta$ have absolutely summable Fourier series with the same sums, and $|\hat{g}_n| \leq \langle|g|\rangle$. With $\langle h|\sigma^2 G|\rangle \leq \langle h(\sigma^2 G)^2\rangle^{1/2}$ and (BE-2), the remainders a_5 and a_6 become proportional to the ripple. In the cells of certificate (ii) this lowers the reachable τ. Single columns close for $\tau \geq 8$ at $\ell \in [55.26, 65.26]$, for $\tau \geq 10$ at $\ell \in [215.26, 225.26]$, and for $\tau \geq 12$ at $\ell \in [455.26, 465.26]$, $[695.26, 705.26]$ and $[855.26, 865.26]$ [proof, computer-assisted: `ci_link.py`, single columns]. A full sweep for $\tau \geq 12$ was not completed and is not used in Theorem 8.

B. The range of σ

The full bound is certified for $\ell \leq 960.8$ (certificate (ii)); a column near $\ell = 950$ needs 8.2×10^7 boxes. The main reduced inequality alone is certified up to $\ell = 6907.8$ (certificate (i)). A relative bound of the remainders by the main numerator would carry the full bound to that range; the available margin is $(T_{10} - T_0)/T_{10} \geq 1.4 \times 10^{-3}$. Point searches indicate that beyond $\ell \approx 3500$ the old remainder a_6 needs $\tau \geq 30$ [numerical]; Lemma A* removes the window-generic γ_P from a_6. The limit $\sigma \to 0$ cannot be reached by finitely many cells; it needs an analytic lemma for large ℓ, as in Corollary 13 of [1].

C. Which windows have coefficient one

For a general window the theorem keeps the factor λ/Υ. Coefficient one is proved for the windows (i)–(iii) of Theorem 8. A possible direction [conjecture]: if λ/Υ could be bounded through the concentration of the spectrum, for example through the cumulative share $F_N = \sum_{n \leq N} n^2 q 0_n^2 / S_2$, the seventeen window-by-window certificates might be replaced by one sufficient condition.

Conjecture C is false. In earlier work we used the working hypothesis $\eta e^{\mu/2} \leq C$, with $C \approx 1$. Explicit windows refute it (App. B). Each is a Rudin–Shapiro band plus a wide coherent block of about 10^8 to 10^{70} harmonics with a tiny raw amplitude. They reach $\log_{10}(\eta e^{\mu/2}) = 32.0$ at $\sigma = 10^{-300}$. The reason is that η adds up coefficients (an ℓ^1 quantity) while μ is set by energies (an ℓ^2 quantity). The weak form $\mu \leq 2\ln(1/\eta) + O(\ln\ln)$ is false as well. The coefficient one is not affected: the upper value of $g\sqrt{\ell}$ on these windows is 1.12 to 2.14, above the band alone, and Theorem 8(ii) covers them.

D. Hypotheses on the window class

- The constant $\bar{\varepsilon}$ of (P6) is not closed for band windows; the third-order terms are open [conjecture].
- (P2) is used through Lemma Z3′. Without (P2) the bound fails already for period averages: windows without a dominant fundamental (beats, combs) exceed $\Pi^*(\mu)$ by factors 3.48 to 318 [numerical]. Whether a weaker hypothesis suffices is open.
- $x \geq 5 \times 10^{11} \sigma^2$ makes $\beta \leq 10^{-6}\tau$ and gives $\tau \geq 5 \times 10^7$ on the small-b edge; it is a technical condition.

E. The upper side

This paper proves only the lower side. The upper side ($g\sqrt{\ell} \leq 1 + \dots$) for general windows has not been attempted; for the windows of [1] it is proved there (Theorem 12 and Corollary 13 of [1]).

Picture. The nearest restriction is the narrow focus. There the observer sees the ripple's brightness locally. Positivity alone makes local brightness pay for fineness (Lemma 9), a ripple hidden at the centre of the focus is dim (Lemma 10), and keeping the local averages costs almost nothing (Lemma 12). What is missing is the cell proof that turns the numerical envelope into a theorem.

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The author used AI assistants in this work. Anthropic Claude (claude-opus-5-5) was used in two separate sessions. In the first session it planned the study and wrote the instructions for each stage, re-ran every verification

suite and re-checked a column of the certificates independently, pointed out that the main theorem had to cover every state of a window (and not only the states near the minimum of the envelope) before it could be stated for $g_{\min}$, found that a sentence of the draft misstated the scope of paper F, and reviewed and edited the manuscript. In the second session it derived the lemmas (Lemma R, Lemma A4*, Lemma H, the split sandwich, the x-free envelope, Lemma BE and the local corrections of Sec. IX A), wrote the cell-based computer-assisted certificates, constructed the counterexamples to conjecture C and the thin-block windows, wrote the evaluation and verification code, and drafted the text; it also found and reported weaknesses of its own and of the first session's reasoning, among them that a set of numbers used as evidence for the narrow-focus states was $\sqrt{\ell}$ itself, and corrected the rounding direction of several constants. While writing the specification of the cells, the second session found that the original acceptance rule did not by itself cover floating-point rounding at large ℓ; it derived the rounding bounds of App. E and recomputed every certificate with the new rule, and all of them closed again. Results were passed between the sessions as written summaries and checked in the first session. Google Gemini (Gemini 4 Argon) and OpenAI ChatGPT (GPT-5.6 Luna) independently reviewed the manuscript, re-deriving the main steps of Lemmas A4*, S, S′ and BE; their comments led to the line-by-line proofs of App. D, the specification of the computer-assisted proofs in App. E, the restriction of Lemma Z3′ to $\mu \geq 2$, references to paper F by its published numbering, and several corrections of notation. For version 1.1, the second session continued with Anthropic Claude Fable (claude-fable-5-1) for the work on the narrow-focus states and returned to claude-opus-5-5 for the repair of the certificates. While trying to close the narrow focus it found a counterexample to the local Z3′ conjectured in v1.0, proved Lemmas CM-1, CN-1, CN-2 and R_θ, and, while building cells for them, found that the boxes of v1.0 did not cover a wedge of states inside each column of certificate (ii); a covering audit then found two further gaps (the extreme values of μ and the smallest b of certificate (i)). It closed all three with new boxes and wrote the random covering audit. It also found and corrected two of its own intermediate claims about the narrow focus. The first session re-ran every verification suite, recomputed the heaviest new columns independently with identical box counts, checked the covering argument line by line, and decided the order of the work. Google Gemini (Gemini 4 Argon) and OpenAI ChatGPT (GPT-5.6 Luna) independently reviewed the changed parts of version 1.1, re-deriving the covering argument, the skip rule and the corner argument for the μ tails and the narrow-focus lemmas; their comments led to the statement of the floating-point library and its measured errors in App. E and the current script names in App. D. For version 1.2, the second session, working on the narrow-focus states with claude-opus-5-5, found that the lower end point in G of the Lemma R* tables is the limit $G \to 0^+$, located the one term responsible by a line-by-line audit of the term tables, and re-ran every certificate of Theorem H1 with this limit by wrapping the published code. Continuing with Anthropic Claude Fable (claude-fable-5-2), it found that the scaling step of Sec. IV used $-N \geq -\lambda N$, derived the identity $T(\lambda) = \lambda T(1) + (1-\lambda)(P_{\mathrm{bg}} - N)$, and checked it numerically at random states; back with claude-opus-5-5, it wrote the proof that the two ends $\lambda = 1$ and $\lambda = 2/3$ suffice, the wrapper of the second certificate with its equivalence and power checks, and ran its first 43 columns. The first session checked both corrections against the published source, recorded them as known issues in the repository as soon as they were found, ran the remaining 233 columns of the second certificate, re-ran every verification suite, and assembled this version. The author directed the research, made all decisions on its content, and takes full responsibility for this article.

Data and code availability. All code, stored results, the computer-assisted certificates and the verification suite are available at `https://github.com/neginyan/resolution-gap-weak-ripple` and archived at Zenodo, doi:10.5281/zenodo.23932577. A single entry point, `verification/run_all.py`, re-derives every number of this paper from the stored results and recomputes a sample of every certificate with the rounding-covering acceptance rule of App. E (about twenty minutes on one CPU core); `--recompute` recomputes all of them. Paper F and its repository [1] are not changed.

Appendix A: The τ factors and Lemma H

Lemma 13 (Lemma H)**.** *[proof]Split $h - 1 = e^{-w^2/2s^2}(R_{\mathrm{lo}} + R_{\mathrm{hi}})$ at frequency m. For the kernels $K = 2\sum \kappa_n \cos nt$ of App. D of [1],*

$$|\langle K(h-1)\rangle| \leq \eta_{\mathrm{lo}}\|K\|_2 + \eta_{\mathrm{hi}}\|K_{\geq m}\|_2. \tag{A1}$$

Since $\kappa_n \propto n^k e^{-n^2\tau/2+n\beta}$, the high part carries $e^{-m^2\tau/2}$, and $m^2\tau = u^2/b \geq u^2$.

τ cells. [proof, computer-assisted: `cb_tau.py`] The τ-dependent factors of the bound decrease in τ and increase in β. On 400 cells of $[20, 80]$, with $\beta \leq 10^{-6}\tau$ and an audit of 20000 points, their product is at least 0.99977. On $[40, 80]$ it is at least $1 - 1.1 \times 10^{-8}$. In certificate (ii) of Sec. VI the ν's are used directly at $\tau = 20$. This is valid on the whole range $\tau \geq 20$ [proof]. Indeed $\tau^j \nu_k(\tau, 10^{-6}\tau) = 2\sum_n n^k \tau^j e^{-c_n\tau}$ with $c_n = n^2/2 - 10^{-6}n \geq 1/2 - 10^{-6}$, and

$$\frac{d}{d\tau}\,\tau^j e^{-c\tau} = \tau^{j-1}e^{-c\tau}\,(j - c\tau) < 0 \quad \text{for } \tau > j/c. \tag{A2}$$

Since $j \leq 2$, $j/c_n < 4.0001 < 20$, so every term decreases on $\tau \geq 20$. For $\omega_2 = 2\sum_n e^{-n^2\tau/2}(n^2\tau - 1)$ (here $n^2\tau \geq 20$) each term has the derivative $(n^2/2)e^{-n^2\tau/2}(3-$

$n^2\tau) < 0$. A 60-digit evaluation on 4000 points of $[20, 10^8]$ finds no increase [numerical]; with $\beta = 0.6\tau$ in place of $10^{-6}\tau$ (so that $c_1 < 0$) it finds an increase at every point (`cq_checks.py`).

Appendix B: Windows

Rudin–Shapiro bands. $f = 1 + c\cos t + c\sum_{k<W} \epsilon_k \cos((m+k)t)$ with the Rudin–Shapiro signs ϵ_k, $W = 2^j$. Non-negativity follows from $|P_j|^2+|Q_j|^2 = 2W$. The blurred sup is bounded by Abel summation with the partial sums $(2+\sqrt{2})\sqrt{n}$.

Counterexamples to conjecture C. Add a coherent block (all phases zero) of N_c harmonics at $u_c = 0.3$ with raw amplitude A. Non-negativity follows from $1 - c(1+\sqrt{2W}) - A(1+N_c) \geq 0$; (P2) and (P3) hold by construction.

σ	η	μ	$\log_{10}\eta e^{\mu/2}$	upper $g\sqrt{\ell}$
10^{-48}	0.027	15.15	1.7	1.492
10^{-96}	0.071	38.08	7.1	1.423
10^{-300}	0.027	154.8	32.0	1.120
10^{-300}	0.071	149.6	31.3	1.248
10^{-300}	0.111	134.7	28.3	1.427

Search for counterexamples to Lemma S. The search draws gapped and gap-free windows with 1 to 300 harmonics on each side, coherent or random, and hill-climbs on the worst margin. We used 300 windows of each kind; the smallest margins were 6.7×10^{-3} and 1.3×10^{-2}.

Appendix C: Verification

`verification/run_all.py` recomputes a sample of every certificate in less than thirty minutes; `--recompute` recomputes all of them. It also re-derives every number of the manuscript from `results/` and checks that it appears in the text. Lemma BE is checked by `ch_bedge.py`; the local corrections of Sec. IX A by `ci_rtheta.py`; the single columns with Lemma A* by `ci_link.py`. The covering of Sec. VI is checked by `cq_cover.py` (10^5 random states and eleven damaged tables, in the default run); sample columns of the μ tails and of the wedge are recomputed by `cq_core.py`, whose term functions are regenerated by `cq_make.py` and compared with the stored ones; the new steps are checked by `cp_checks.py` and `cq_checks.py`. The results on the narrow focus (Sec. IX A, App. F) are produced by `cm_counter.py`, `cm_local_bound.py`, `cm_checks.py`, `cn_lemmas.py`, `cn_search.py`, `co_remainders.py`, `co_rtheta.py` and `co_envmin.py`. The acceptance rules of App. E are implemented in `cl_cells.py` and `cl_link.py`; the bounds are checked against long double by `ck_rounding.py`, and the τ cells and Lemma BE by `cl_tau_be.py`. Each step of App. D is checked independently, in exact fractions and 60-digit decimal, by `ck_details.py`. The stage scripts are listed with their outputs in the repository README.

Appendix D: Detailed proofs

This appendix writes the new lemmas out line by line. Each step has an independent check in `verification/ck_details.py` (exact rational arithmetic, Fourier sums, random states, and 60-digit decimal constants); it runs in the default of `run_all.py`.

1. Conventions

Functions are 2π-periodic in t and depend on w. For such a function $u(t,w)$ and a positive measure m on the w-axis we write

$$\langle u\rangle_t = \frac{1}{2\pi}\int_0^{2\pi} u\,dt, \qquad \langle u\rangle = \int \langle u\rangle_t\, dm(w), \tag{D1}$$

$$\|y\|_t^2 = \langle y^2\rangle_t, \qquad \|u\|^2 = \langle u^2\rangle. \tag{D2}$$

So $\langle\cdot\rangle_t$ is the normalised period average, and $\|\cdot\|$ is the L^2 norm of $dt/2\pi \otimes m$. Fourier coefficients are $\hat{u}_n = \langle u\, e^{-int}\rangle_t$, so $u = \sum_n \hat{u}_n e^{int}$ and $\langle KU\rangle_t = \sum_n \hat{K}_n\hat{U}_{-n}$. For $R = \sum_{n\geq 1} q0_n \cos(nt - \varphi_n)$, Parseval gives $\|R\|_t^2 = S_0/2$ and $\|R'\|_t^2 = S_2/2$.

2. Lemma S

Data. $Y = \phi(w)y(t)$ with $y = R$ or $y = R'$, $h = 1 + r$ with $r = \psi(w)R(t)$, $0 \leq \psi \leq 1$, $\eta = \sup|R| < 1$. The split $R = R_{\rm lo} + R_{\rm hi}$ puts the frequencies $\leq M$ in $R_{\rm lo}$ and those $\geq m$ in $R_{\rm hi}$, $M < m$. Then $y = y_{\rm lo} + y_{\rm hi}$, $Y = Y_{\rm lo}+Y_{\rm hi}$, $r = r_{\rm lo}+r_{\rm hi}$, with $|r_{\rm lo}| \leq \eta_{\rm lo}$ and $|r_{\rm hi}| \leq \eta_{\rm hi}$.

Step 1 (orthogonality). $y_{\rm lo}$ and $y_{\rm hi}$ have disjoint frequencies, so $\langle y_{\rm lo}y_{\rm hi}\rangle_t = 0$. Hence $\|Y\|^2 = \|Y_{\rm lo}\|^2 + \|Y_{\rm hi}\|^2$ and $f_{\rm lo} + f_{\rm hi} = 1$.

Step 2 (algebra). For $|r| < 1$,

$$\frac{1}{1+r} = 1 - r + \frac{r^2}{1+r}, \qquad 0 \leq \frac{r^2}{1+r} \leq \frac{r^2}{1-\eta}. \tag{D3}$$

Hence

$$\Big\langle \frac{Y^2}{h}\Big\rangle = \|Y\|^2 - \langle Y^2 r\rangle + \Big\langle \frac{Y^2 r^2}{1+r}\Big\rangle, \tag{D4}$$

with the last term in $[0, \|Yr\|^2/(1-\eta)]$.

Step 3 (six products). Expanding,

$$\begin{aligned}\langle Y^2 r\rangle = {} & \langle Y_{\rm lo}^2 r_{\rm lo}\rangle + \langle Y_{\rm lo}^2 r_{\rm hi}\rangle + 2\langle Y_{\rm lo}Y_{\rm hi}r_{\rm lo}\rangle \\ & + 2\langle Y_{\rm lo}Y_{\rm hi}r_{\rm hi}\rangle + \langle Y_{\rm hi}^2 r_{\rm lo}\rangle + \langle Y_{\rm hi}^2 r_{\rm hi}\rangle.\end{aligned} \tag{D5}$$

(a) $|\langle Y_{\rm lo}^2 r_{\rm lo}\rangle| \leq \eta_{\rm lo}\|Y_{\rm lo}\|^2$. (b) With the gap $2M < m$: $y_{\rm lo}^2$ has frequencies $\leq 2M$ and $R_{\rm hi}$ frequencies $\geq m$,

so $\langle y_{\rm lo}^2 R_{\rm hi}\rangle_t = 0$ and $\langle Y_{\rm lo}^2 r_{\rm hi}\rangle = 0$. (c) With the gap: $y_{\rm lo}R_{\rm lo}$ has frequencies $\le 2M$, $y_{\rm hi}$ frequencies $\ge m$, so $\langle Y_{\rm lo}Y_{\rm hi}r_{\rm lo}\rangle = 0$. (d) By Cauchy–Schwarz in $\langle\cdot\rangle$, $|\langle Y_{\rm lo}Y_{\rm hi}r_{\rm hi}\rangle| \le \eta_{\rm hi}\|Y_{\rm lo}\|\|Y_{\rm hi}\|$. (e) Two bounds hold. First, $|\langle Y_{\rm hi}^2 r_{\rm lo}\rangle| \le \eta_{\rm lo}\|Y_{\rm hi}\|^2$. Second, with $0 \le \psi \le 1$ and Cauchy–Schwarz in t,

$$|\langle Y_{\rm hi}^2 r_{\rm lo}\rangle| \le \int \phi^2\, dm\, \langle y_{\rm hi}^2 |R_{\rm lo}|\rangle_t \le \int \phi^2\, dm\, \sup|y_{\rm hi}|\, \|y_{\rm hi}\|_t \|R_{\rm lo}\|_t. \quad \text{(D6)}$$

Since $\|Y_{\rm hi}\|^2 = \int \phi^2 dm\, \|y_{\rm hi}\|_t^2$, the right side is $\|Y_{\rm hi}\|^2 \sup|y_{\rm hi}|\, \|R_{\rm lo}\|_t/\|y_{\rm hi}\|_t$. Together: $|\langle Y_{\rm hi}^2 r_{\rm lo}\rangle| \le \kappa\|Y_{\rm hi}\|^2$. (f) $|\langle Y_{\rm hi}^2 r_{\rm hi}\rangle| \le \eta_{\rm hi}\|Y_{\rm hi}\|^2$. Dividing by $\|Y\|^2$ gives $|\langle Y^2 r\rangle| \le E\|Y\|^2$.

Step 4 (the square). Write $Yr = Y_{\rm lo}r_{\rm lo} + Y_{\rm hi}r_{\rm lo} + Yr_{\rm hi}$. By Minkowski,

$$\|Yr\| \le \|Y_{\rm lo}r_{\rm lo}\| + \|Y_{\rm hi}r_{\rm lo}\| + \|Yr_{\rm hi}\|. \quad \text{(D7)}$$

Here $\|Y_{\rm lo}r_{\rm lo}\| \le \eta_{\rm lo}\sqrt{f_{\rm lo}}\|Y\|$ and $\|Yr_{\rm hi}\| \le \eta_{\rm hi}\|Y\|$. The middle term uses the same two bounds as (e): $\|y_{\rm hi}R_{\rm lo}\|_t \le \eta_{\rm lo}\|y_{\rm hi}\|_t$ and $\|y_{\rm hi}R_{\rm lo}\|_t \le \sup|y_{\rm hi}|\,\|R_{\rm lo}\|_t$, so $\|Y_{\rm hi}r_{\rm lo}\| \le \kappa\sqrt{f_{\rm hi}}\|Y\|$. Hence $\|Yr\|^2 \le B\|Y\|^2$; the same κ enters E and B.

Step 5 (Lemma S). Steps 2–4 give $\|Y\|^2(1-E) \le \langle Y^2/h\rangle \le \|Y\|^2(1+E+B/(1-\eta))$. The η sandwich $1-\eta \le h \le 1+\eta$ gives the other two bounds; take the better of each pair.

Lemma S′. Without the gap, (b) and (c) are bounded instead of vanishing: $|\langle Y_{\rm lo}^2 r_{\rm hi}\rangle| \le \eta_{\rm hi}\|Y_{\rm lo}\|^2$ and $2|\langle Y_{\rm lo}Y_{\rm hi}r_{\rm lo}\rangle| \le 2\eta_{\rm lo}\|Y_{\rm lo}\|\|Y_{\rm hi}\|$. These add $\eta_{\rm hi}f_{\rm lo} + 2\eta_{\rm lo}\sqrt{f_{\rm lo}f_{\rm hi}}$ to E. Step 1 needs only $M < m$, and Step 4 uses no gap, so B is unchanged.

Where Lemma S is used. Lemma S is a statement about period averages. In the windows of this paper the v-average is Gaussian, i.e. the period average weighted by θ_τ (Lemma 19 of [1]). Lemma R first converts every Gaussian average into a period average, with the explicit ν factors of Lemmas 21 and 22 of [1]: $J_{ww} = J_{\rm per}(1+\epsilon_1)$, $|\epsilon_1| \le \nu_0(1+2\bar{Q})/(1-z)$, and the two-sided bound of W by $W_{\rm per}$ with $(1\pm\nu_0)$ and $\pm\frac12 b\tau\bar{Q}\nu_2$. Only then is a sandwich applied, to three period averages, each of the form of Lemma S with $\psi = E = e^{-w^2/2s^2}$:

- $J_{{\rm per},R} = \int \Gamma\langle h_w^2/h\rangle$: $y = R$, $\phi = -(w/s^2)E$, $dm = \Gamma\, dw$;
- $W_{\rm per} = x\int \Gamma\langle h_t^2/h\rangle$: $y = R'$, $\phi = E$, $dm = x\,\Gamma\, dw$;
- the F term of (R4): $y = R'$, $\phi = E$, $dm = x\,\Gamma(1-\cos\frac{w}{\sqrt2})\, dw$, a positive measure.

Lemma S replaces (R1) in each of them, and nothing else changes. The ν factors are those of certificate (ii), taken at $\tau = 20$. Lemma H (App. A) is not used in this chain.

3. Lemma A4*

Kernels. With $e_n = e^{-n^2\tau/2}$, $\theta_\tau = \sum_n e_{|n|}e^{int}$ and $\Theta = \sum_n e_{|n|}e^{n\beta}e^{int}$ (Lemma 19 of [1]). So $(\theta_\tau - \Theta)\hat{}_n = e_{|n|}(1-e^{n\beta})$, and for any (complex) g,

$$\langle(\theta_\tau-\Theta)g\rangle_t = \sum_{n\ge1} e_n\big[(1-e^{n\beta})\hat{g}_{-n} + (1-e^{-n\beta})\hat{g}_n\big]. \quad \text{(D8)}$$

With $e^{\pm n\beta} = \cosh n\beta \pm \sinh n\beta$ this is

$$\sum_{n\ge1} e_n\big[(\hat{g}_n+\hat{g}_{-n})(1-\cosh n\beta) + (\hat{g}_n-\hat{g}_{-n})\sinh n\beta\big]. \quad \text{(D9)}$$

This identity does not need g to be real. Here $g = Fe^{i\psi}$ is complex, and $|\hat{g}_{\pm n}| \le \langle|g|\rangle_t = \langle F\rangle_t$ because $F \ge 0$.

First sum. $\cosh u - 1 \le \frac{u^2}{2}e^u$ gives at most $\sum_n e_n\, 2\langle F\rangle_t \frac{n^2\beta^2}{2}e^{n\beta} = \frac{\beta^2\nu_2}{2}\langle F\rangle_t$, with $\nu_k = 2\sum n^k e_n e^{n\beta}$.

Second sum. Write $\psi = w/\sqrt2 + \chi$ and $g = g^0 + g^1$, $g^0 = e^{iw/\sqrt2}F$, $g^1 = e^{iw/\sqrt2}F(e^{i\chi}-1)$. F is real, so $\hat{F}_{-n} = \overline{\hat{F}_n}$ and $\hat{g}^0_n - \hat{g}^0_{-n} = 2i\, e^{iw/\sqrt2}\,{\rm Im}\hat{F}_n$. Its real part is $-2\sin(w/\sqrt2)\,{\rm Im}\hat{F}_n(w)$.

Parity. In every state $h = 1 + e^{-w^2/2s^2}R(t)$, so h, h_t and $F = k'^2h_t^2/h$ are even in w, and so is ${\rm Im}\hat{F}_n(w)$. Γ is centred, e and β do not depend on w, so

$$\int \Gamma(w)\, \sin\frac{w}{\sqrt2}\, {\rm Im}\hat{F}_n(w)\, dw = 0 \quad \text{(D10)}$$

for every n and every allowed state; the g^0 part of the second sum vanishes after integration in w.

The rest. $|g^1| \le F|\chi| \le F\bar{\chi}(w)$, so $|\hat{g}^1_n - \hat{g}^1_{-n}| \le 2\bar{\chi}\langle F\rangle_t$, and $\sinh u \le ue^u$ gives at most $\beta\nu_1\bar{\chi}(w)\langle F\rangle_t$.

The pieces of $\bar{\chi}$. $\chi = \sigma^2(\tilde{s}+s_w)/\sqrt2$ with $\sigma^2|\tilde{s}| = \sigma^2k'|h_t|/h \le \xi = \sigma\sqrt{x}N_1/(1-\bar{Q})$ and $\sigma^2|s_w| \le \sigma^2|w|/L^2 + \zeta$, $\zeta = \sigma^2\eta e^{-1/2}/(s(1-\eta))$ (from $\sup_w |w|e^{-w^2/2s^2}/s^2 = e^{-1/2}/s$). So $\bar{\chi} = (\xi + \zeta + \sigma^2|w|/L^2)/\sqrt2$.

Integration in w. $e\sigma^2\int\Gamma\langle F\rangle_t = eW_{\rm per}$. The weight $\langle F\rangle_t \propto e^{-w^2/s^2}$ turns Γ into a centred Gaussian of variance s_{e2}, whose mean of $|w|$ is $\sqrt{2s_{e2}/\pi}$; so the $|w|$ part gives $\sigma^2\sqrt{s_{e2}/\pi}/L^2$. The η sandwich bounds $W_{\rm per}$ by $W_{\rm per,up}$. This is a_4^*. Every piece is $W_{\rm per}$ times $\beta^2\nu_2$ or $\beta\nu_1\sigma$: second order in β, or first order in β and in σ.

4. Lemma Z3′

Setting. (P2) and $K \ge x$ give $q0_n \le q0_1 d_n$, $d_n = e^{-(n^2-1)x/2}$; (P3) gives $\sum q0_n \le \bar{Q}$. Put $y = q0_1$, $v_n = q0_n/y \in [0, d_n]$ $(n \ge 2)$, $s = \sum v_n^2$, $t = \sum v_n$. Then

$S_0 = y^2(1+s)$, $P = \mu S_0 = \mu y^2(1+s)$ and $y(1+t) \le \bar{Q}$, so

$$P \le \bar{Q}^2 \mu \, \frac{1+s}{(1+t)^2}. \tag{D11}$$

(i) Mean identity. $\mu(1+s) = x(1+\sum n^2 v_n^2)$, i.e. $\sum_{n\ge 2} v_n^2 (n^2 x - \mu) = \mu - x$. On $H = \{n^2 x > \mu\}$, $v_n^2(n^2x - \mu) \le v_n c_n$ with $c_n = d_n(n^2x-\mu) = e^{(x-\mu)/2} z e^{-z/2} \le \frac{2}{e} e^{(x-\mu)/2}$, $z = n^2 x - \mu$.

(ii) $s \le a\,t$ with $a = c_s(x) e^{-\mu/2}$, $c_s(x) = e^{x/2}(1 + \sqrt{e} + 2/e)$. On H, $v_n^2 \le d_n v_n \le e^{(x-\mu)/2} v_n$. On $\{0 \le \mu - n^2 x \le 1\}$, $v_n \le d_n \le e^{(x-\mu+1)/2}$. On $\{\mu - n^2 x > 1\}$, $\sum v_n^2 \le \sum v_n^2 (\mu - n^2 x) \le \sum_H v_n c_n \le \frac{2}{e} e^{(x-\mu)/2} t$ by (i).

(iii) By (i), $\mu - x \le \sum_H v_n c_n \le \frac{2}{e} e^{(x-\mu)/2} t$, so $t \ge t_0(x) = \frac{e}{2}(\mu - x) e^{(\mu - x)/2}$.

(iv) Monotonicity. $\frac{d}{dt} \frac{1+at}{(1+t)^2} = \frac{a-2-at}{(1+t)^3}$. If $a \le 2$ this is negative for every $t \ge 0$; so $\frac{1+s}{(1+t)^2} \le \frac{1+at}{(1+t)^2} \le \frac{1+at_0}{(1+t_0)^2}$.

Removing x. For $x \le 0.003 < \mu$, $t_0(x)$ decreases in x (it is increasing in $\mu - x$), so $t_0(x) \ge t_0(0.003)$. $c_s(x)$ increases in x, so $a \le c_s(0.003) e^{-\mu/2}$; and the bound of (iv) increases in a. This gives $\Pi^*(\mu)$, with $c_s = c_s(0.003) = 3.3896$. Finally $s \le t$ (as $v_n \le 1$), so $(1+s)/(1+t)^2 \le 1$ and $P \le \mu \bar{Q}^2$.

Range. (iv) needs $a \le 2$, i.e. $\mu \ge 2\ln(c_s/2) = 1.055$. The certificates use $\Pi^*(\mu)$ only for $\mu \ge 2$ and the bound $\mu \bar{Q}^2$ for $\mu < 2$ (`cc_cells.py` and `cd_cells.py` with the rule of `cl_cells.py`, and `cl_link.py`). For $\mu \in (0.003, 0.226)$ the formula Π^* is not a valid bound by this argument, because $t_0 < (a-2)/a$ there; the statement in Sec. V is restricted accordingly. The envelope values of Sec. V are attained at $\mu > 5$.

5. Lemma BE

Where the cross term enters. In the exact identity of Lemma 23 of [1], the cross term is $\sigma^2 \mathrm{Re} A[h\, 2\alpha\psi' e^{i\psi}]$ with $2\alpha\psi' = (1-b)\sigma^2 G$. After the θ_τ steps (remainders a_5, a_6) it is $e(1-b)\,\sigma^2 \int \Gamma \langle h\, \sigma^2 G \cos\psi \rangle$, divided by $Z \ge 1 - z$. The factor σ^2 in front is the one that produces $-\sigma^2 \gamma_P \Phi$ in Lemma 23 of [1].

(BE-1) applied. With (BE-1) and $e(1-b) \le e$, this term is at least

$$-\, e(1-b)\sigma^4 \int \Gamma \langle F \rangle \\ - e\sigma^2 \Big[\frac{M_G}{4c} + 2c\,\Phi \Big]. \tag{D12}$$

The first part is the $-\sigma^2 W$ of Lemma 23 of [1]; the bracket replaces $\gamma_P \Phi$.

The choice of c. With $c = (b + \sigma^2 J_{ww})/(16\sigma^2)$:

$$e\sigma^2 \cdot 2c\,\Phi = \frac{e(b+\sigma^2 J_{ww})\Phi}{8} \le \frac{b\Psi}{2} + \frac{\sigma^2 J_{ww}}{4}, \tag{D13}$$

by $\Phi \le 4\Psi$ (Lemma 23 of [1]) and $\Phi \le 2$; and

$$e\sigma^2 \frac{M_G}{4c} = \frac{4e\sigma^4 M_G}{b + \sigma^2 J_{ww}}. \tag{D14}$$

The M_G part. By (BE-2) with $x^2 S_4 \le \Lambda x S_2 + T_\Lambda$,

$$M_G \le \frac{3 g_s}{2} \Big[x S_2 K_1 + \frac{T_\Lambda}{1 - \bar{Q}} \Big]. \tag{D15}$$

Ripple piece. Use $x S_2 g_s \le 2(1+\bar{Q}) W_{\mathrm{per}}$ and $b + \sigma^2 J_{ww} \ge b$:

$$\frac{4e\sigma^4}{b} \cdot \frac{3}{2} \cdot 2(1+\bar{Q}) K_1 W_{\mathrm{per}} = \frac{12(1+\bar{Q}) e \sigma^4 K_1}{b} W_{\mathrm{per}}. \tag{D16}$$

With $1 - e \ge y/(2+y)$, $y = (1-b)^2\sigma^2/(2b)$, i.e. $\sigma^2/(by) = 2/(1-b)^2$,

$$\frac{12(1+\bar{Q}) e \sigma^4 K_1}{b(1-e)} \le 12(1+\bar{Q}) K_1 \sigma^2 \frac{2}{(1-b)^2} \\ \times e^{-y/2}(2+y) \le \frac{48(1+\bar{Q}) K_1 \sigma^2}{(1-b)^2} = \rho_W, \tag{D17}$$

since $e^{-y/2}(2+y) \le 2$. With $\Lambda = 10^{-3}/\sigma^2$, $\sigma^2 K_1 = 10^{-3}/(1-\bar{Q}) + \sigma^2 (U_1^2 + \frac{1}{2})/(1-\bar{Q})^3$. The second term is below 10^{-40} for $\sigma \le 10^{-24}$ and $U_1 \le 29.4$, so $\rho_W = 0.08 + O(10^{-40}) \le 0.0801$.

Tail piece. Use $b + \sigma^2 J_{ww} \ge \sigma^2 J_{ww}$:

$$\frac{4e\sigma^4}{\sigma^2 J_{ww}} \cdot \frac{3 g_s}{2} \cdot \frac{T_\Lambda}{1 - \bar{Q}} \le \frac{6\sigma^2 g_s T_\Lambda}{(1-\bar{Q}) J_{ww}}. \tag{D18}$$

The factor σ^2 stays: it is the σ^2 of the cross term times $\sigma^4/(\sigma^2 J_{ww})$. We compare it with $\sigma^2 J_{ww}$, where $J_{ww} \ge 0.6/L^2 \ge 0.6\sigma$: the ratio is $8 g_s T_\Lambda / J_{ww}^2 \le 22.3\, T_\Lambda/\sigma^2$. With $T_\Lambda \le 4e^{-\Lambda}(\Lambda^2 + \Lambda^{3/2} x^{-1/2}/1.25)$ and $x^{-1/2} \le 1/(7 \times 10^5 \sigma)$ the ratio is below $10^{-4 \times 10^{44}}$.

Collecting. Lemma 23 of [1], with this replacement and $\Psi \ge \min(\Phi, \frac{1}{2})$, gives

$$2N_\phi \ge (1-e) W \Big(1 - \frac{\sigma^2}{(1-e)(1-\nu_0)} \\ - \rho_W \frac{1+z}{(1-z)(1-\nu_0)} \Big) \\ + b\Psi \Big(\frac{1}{1+z} - \frac{1}{2(1-z)} \Big) - \frac{\sigma^2 J_{ww}}{4(1-z)} - \mathrm{rem}. \tag{D19}$$

Here $W_{\mathrm{per}} \le ((1+z)W + \frac{1}{2} b \tau \bar{Q} \nu_2)/(1-\nu_0)$ was used, and $\sigma^2/(1-e) \le \sigma^2 + 4b/(1-b)^2 \le 4.01 \times 10^4 \sigma^2$. With $2A_w \ge \sigma^2 J_{ww}$, $(a) + (a') = (2A_w + N_\phi)/(\sigma\,\mathrm{den})$, and $\nu_0, z \le 10^{-10^7}$, this is the inequality of (BE-3).

Remainders. rem collects $a_1, a_2, a_3, a_5, \ldots, a_{10}$, a_4^*, $\frac{1}{2} b \bar{Q}(\nu_0 + \tau\nu_2)\Phi$, the ν part of the W sandwich and the tail piece. Each is a product of a power of $1/\sigma$ (at most σ^{-2} against $0.6\sigma^3$), a power of $\tau \le$ any, and $\nu_k \le 2.0000001 e^{-0.499999\tau}$ with $\tau \ge 5 \times 10^7$. $\tau^j e^{-\tau/2}$ decreases there, so the worst end is $\tau = 5 \times 10^7$, $\sigma = 10^{-3000}$, and each ratio is below $10^{-1.08 \times 10^7}$.

Rounding. Every printed constant is rounded in the safe direction: κ and 1.14×10^{-5} down (value 1.1500×10^{-5}), $1 - e$ and y down, Ψ down (0.22120 to 0.2211), ρ_W up.

Appendix E: Specification of the cell proofs

This appendix states what a reader needs to check that the cell proofs of Sec. VI are sound: the floating-point model, the rounding bounds, the acceptance rule, the monotonicity of every term and the treatment of underflow. The cells are evaluated in double precision, not in interval arithmetic; soundness comes from an acceptance margin that is larger than a derived bound of the rounding error.

Floating-point model. IEEE 754 binary64 with round to nearest, unit roundoff $u = 2^{-53}$. We assume that log, exp, $\log 1p$, expm1 and $\sqrt{}$ return a value within two ulp of the exact value at the floating argument (relative error $\leq 4u$). IEEE 754 guarantees this only for $\sqrt{}$ (correctly rounded); for the transcendental functions it is a property of the library. The certificates were run with NumPy 2.5.3 on x86-64 Linux; against a 60-digit decimal evaluation on 2×10^4 to 4×10^4 arguments per function spread over the ranges used, the largest errors were 0.71 ulp (exp), 0.50 (log), 0.59 ($\log 1p$), 0.52 (expm1) and 0.50 ($\sqrt{}$) [numerical] (`ulp_check.py`). The safety factor $S = 10$ below also leaves room for a library whose errors are several times larger than assumed. Cell ends and ℓ are floating numbers and are exact; a cell is the set of real points between its ends, and a bisection reuses the same floating midpoint on both sides, so the cells of a column cover its domain exactly.

Certificate (i): the main reduced inequality. The test compares two logarithms, $\ell_{\rm pos}$ and $\ell_{\rm neg}$, obtained from the cell ends by a chain of at most 25 operations. Each operation has error at most u (or $2u$ for an elementary function) times the size of its operands. The operands are the cell ends (each used at most twice), ℓ, constants below 100, and $\log G_{\rm cell}$, which enters only inside a log-add-exp and is either dominant or dropped exactly. Hence, with $B = 2(|\log r_a| + |\log r_b| + |\log b_a| + |\log b_b| + |\log \mu_a| + |\log \mu_b|) + \ell_b + 100$,

$$|\hat{\ell}_{\rm pos} - \ell_{\rm pos}| + |\hat{\ell}_{\rm neg} - \ell_{\rm neg}| \leq \delta = 60u\,(|\hat{\ell}_{\rm pos}| + |\hat{\ell}_{\rm neg}| + B). \tag{E1}$$

A cell is accepted if $\hat{\ell}_{\rm pos} \geq \hat{\ell}_{\rm neg} + \max(10^{-12}, S\delta)$ with the safety factor $S = 10$ (`cl_cells.py`).

Certificates (ii) and (iii): the full Lemma R bound.* Every term is e^{l_t}, where l_t is a sum of at most 16 operands: cell ends times coefficients $|c| \leq 2$, powers $p\ell$ with $p \leq 3$, constants, $\log G$ and logarithms of monotone factors. By Higham's bound, $|\hat{l}_t - l_t| \leq \gamma_{k+4} \sum |x_i|$, and $\sum |x_i| \leq |l_t| + 2B$ with $B = 2(\max|\log\mu| + \max|\log r| + \max|\log b|) + 3\ell_b + 600$. The factor $e^{-y/2}$ adds $4u\,y_{\max}B$ (if $y_{\max} \geq 1500$ it underflows to zero and the positive term is dropped), and the later steps (netting at equal powers of σ, the shift by $p\log\sigma$, normalisation) add $4u(|l_t| + B + 3\ell_b)$. So

$$|\hat{l}_t - l_t| \leq \eta_t = 25u\,(|l_t| + 3B + y_c B). \tag{E2}$$

The remaining operations (sums of at most 60 terms, netting, the closed-form minimum in a', square roots) act on quantities normalised by the largest part. Their error is at most $1000u$ times the *unnetted* magnitude, i.e. the sum of the absolute values of all terms. This is the point that a rule of the form "value/magnitude $> 10^{-10}$" after netting does not cover: when equal powers of σ nearly cancel, the netted magnitude can be smaller than the unnetted one by the ratio R ($R \leq 7\times10^4$ was observed).

The rule. Before any other step, every positive term is lowered to $l_t - S\eta_t - 1000Su$ and every negative term is raised to $l_t + S\eta_t + 1000Su$, with $S = 10$ (`cl_link.py`). These perturbed terms bound the exact ones in the safe direction, and the margin $1000Su$ covers the rounding of all later steps relative to the unnetted magnitude. A cell is accepted if the resulting normalised minimum exceeds 10^{-10}. Acceptance therefore implies that the exact lower bound of the cell is positive.

A check of the bounds. Recomputing $\ell_{\rm pos} - \ell_{\rm neg}$ in long double (64-bit mantissa) on random cells gives differences of 6.8×10^{-13} at $\ell = 961$ and 2.7×10^{-12} at $\ell = 6898$, below δ (`ck_rounding.py`) [numerical].

Recomputation. All certificates were recomputed with these rules on exactly the stored columns. All 92 columns of (ii), all 17 windows of (iii) and all 987 columns of (i) close. The cell counts are those of the earlier rule in all but 127 columns, where at most 176 more boxes are needed. In (i) the smallest ratio of the accepted margin to $S\delta$ is 1.00. In (ii), at $\ell = 55.26$ the column still closes with $S = 10^7$.

The τ cells and Lemma BE. The factor of App. A is built from at most 80 exponentials, one sum, a few field operations and one square root, so its relative rounding error is at most $20u$. Its margin 8.6×10^{-6} above 0.99977 is more than 3×10^9 times this bound, and a 50-digit decimal evaluation agrees to within 3.5×10^{-16}. The constants of Lemma BE are computed in 60-digit decimal; there is no double-precision acceptance rule there (`cl_tau_be.py`).

Monotonicity. On a cell, every term is bounded by its value at one corner. The direction is fixed by the following table, where "+" means that the term is evaluated at the end that makes it smallest if it is positive and largest if it is negative.

factor	monotone in, and why
σ^p	monotone in ℓ; after netting at equal σ, positive parts at the end that makes σ^p smallest, negative parts at the other ($p < 0$ occurs only in the β boxes)
$\gamma = T_D/\sqrt{\ell}$	decreasing in ℓ; taken at ℓ_a
$y = (1-b)^2\sigma^2/(2b)$	decreasing in b, increasing in σ
$e^{-y/2}$, $1/(1+w_0)$	decreasing in y; positive terms at $y_{\max}$
$(1-b)^2$	decreasing in b
r, $\sqrt{r/(2+r)}$, $1/(2+r)$	monotone in r (powers and ratios of linear functions)
$\mu^{\pm1}$, G/μ	monotone in μ; in the tails see below
$G_{\text{cell}} = \Pi^*(\mu)g_s(r)$	Π^* decreasing for $\mu \geq 2$ (left end), $\mu\bar{Q}^2$ for $\mu < 2$ (right end); g_s increasing in r
b, $\sqrt{b}$, $1/b$	monotone in b
$\tau^j\nu_k$, ω_2	decreasing in $\tau \geq 20$ (proved, App. A); taken at $\tau_{\min}$
U_1, U_2, γ_P, c_χ	increasing in ℓ; taken at ℓ_b

The end points in G. The main reduced function is concave in G (Sec. V), and for each A the Lemma R* bound is affine in G on $(0, G_{\text{cell}}]$, so its minimum over A is concave there. One remainder bound, the term $\tau\nu_2\bar{Q}/(16(1+z))$ of the constants, enters only on the branch $G > 0$ and does not contain G. The value at $G = 0$ is therefore not the limit of the branch, and the lower end is the limit $G \to 0^+$, in which every other term containing G vanishes. A box is accepted when it passes at $G = 0$, at $G \to 0^+$ and at $G = G_{\text{cell}}$. The two ways of treating the F term (kept with its remainders, or dropped since it is ≥ 0) are different lower bounds of the same quantity. They coincide at $G \to 0^+$, because every term in which they differ carries G. So for each state one of them is positive at both ends, and it is positive in between by concavity. The other options of a box (the branches in a', the split points, the shifts of the H term) are relaxations of one and the same bound, and need no such argument. For the main reduced function the limit $G \to 0^+$ equals the value at $G = 0$: every term with G enters through a log-add-exp, which returns its other argument exactly when $\log G = -\infty$. v1.0 and v1.1 evaluated $G = 0$ and $G = G_{\text{cell}}$ only. Every certificate of Theorem H1 has been re-run with the limit $G \to 0^+$ in addition, by wrapping the published acceptance functions without changing them (`ct_v11_G0plus.py`): the 92 columns of the Lemma R* boxes, the 92 wedge columns and the 92 columns of the μ tails of certificate (ii), the 17 windows of certificate (iii), and the 987 tail columns of certificate (i). Every box passes, and every cell count equals the published one. A random check at intermediate G (log-uniform in $[G_{\text{cell}}e^{-700}, G_{\text{cell}}e^{-0.1}]$, 2.25×10^8 evaluations) finds no negative value. The second certificate of the scaling step (Sec. IV) evaluates all three points in every box.

Underflow and clipping. All terms are carried as logarithms and normalised by the largest part before exponentiation. Exponents are clipped at 700 for positive parts (clipping lowers them) and at 709 for negative parts, and a negative part is never rounded down to zero (a floor of 10^{-300} is added). Underflow of a positive part only drops a non-negative quantity. Underflow of $e^{-y/2}$ drops the positive term that carries it.

The μ tails. The variable μ enters the bounds only through G and G/μ.

- Upper tail $[\mu_{\max}, \infty)$. Since Π^* decreases for $\mu \geq 2$, $G \leq \Pi^*(\mu_{\max})g_s(r)$. The positive term with G/μ (the part $GA/(\mu(2+r))$ of the numerator) is bounded below by 0. The negative terms with G/μ are bounded by $\Pi^*(\mu_{\max})g_s/\mu_{\max}$.
- Lower tail $(0, e^{-30}]$. For $\mu < 2$, $P \leq \mu\bar{Q}^2$, so $G = \mu\rho$ with $\rho \in [0, \bar{Q}^2 g_s]$. For fixed A the bound is affine in G and in G/μ, hence affine in μ at fixed ρ and affine in ρ at fixed μ. Its minimum over A is concave along both axes, so its minimum over the rectangle $(\mu, \rho) \in (0, e^{-30}] \times [0, \bar{Q}^2 g_s]$ is attained at a corner. The corners are $G = 0$; $(G, G/\mu) = (0, \bar{Q}^2 g_s)$, the limit $\mu \to 0$; $(e^{-30}\bar{Q}^2 g_s, \bar{Q}^2 g_s)$; and, on the edge $\rho \to 0^+$, the limit of the branch $G > 0$ at both ends of μ (see the end points in G above). A box is accepted when all of them pass. Bounding the positive G terms at $\mu \to 0$ and the negative ones at e^{-30} separately does not close near $b = 10^4\sigma^2$: the W term and its remainder carry the same G.
- The strip $b \to 0$ of certificate (i). The only term increasing in b is $k_0 = b/(8r)$; the strip $(0, b_{\text{lo}}]$ uses $k_0 \geq 0$.

An infinite end never enters a finite operand: it only produces exact $-\infty$ terms, which the log-sum-exp and the netting drop exactly. The bounds B and η_t above therefore use the finite ends only, and the rules are otherwise unchanged.

The wedge: boxes in (μ, r, β, ℓ). The terms are those of certificate (ii), with $b = \beta\sigma^2$.

- (J1) A term $c\, b^k \sigma^p$ becomes $c\, \beta^k \sigma^{p+2k}$; the powers now range over $-1, \ldots, 3$.
- (J2) $y = (1-b)^2/(2\beta)$, so $y \leq 1/(2\beta_a)$ and $y \geq (1-b_{\text{hi}})^2/(2\beta_b)$, independent of σ.
- (J3) Since $b \leq b_{\max}$, a box with $\beta \geq \beta_a$ contains only $\sigma \leq \sigma_{\text{eff}} = \min(e^{-\ell_a}, \sqrt{b_{\max}/\beta_a})$. A box with $\sigma_{\text{eff}} < e^{-\ell_b}$ is empty.
- (J4) $b \leq b_{\text{hi}} = \min(\beta_b \sigma_{\text{eff}}^2, b_{\max})$.
- (J5) Terms of equal power are combined at equal σ and then bounded at the worse end; for $p < 0$ the ends swap.

- (J6) The b part of the bracket, $\beta\sigma^2/(a'+\sigma)$, increases with σ. Its positive net is bounded at $\sigma_{\rm lo}$, shift included; the larger of this and the v1.0 bound is used.

With β alone, the column $[55.26, 65.26]$ does not close. In the wedge the main balance $2\sqrt{\alpha\beta_H}$ grows like σ, and so does the largest cost, $\approx 1.1\sigma$; they are measured at opposite ends of the column, a loss of $e^{\ell_b-\ell_a}$. A box of width 1 in ℓ around the stuck point passes with margin 0.999. Therefore ℓ is a fourth box variable. Each box takes $\gamma = T_{10}(\ell_a)/\sqrt{\ell_a}$, decreasing in ℓ, at its own ℓ_a; the window constants $U_1, U_2, \gamma_P, c_\chi$, increasing in ℓ, at its own ℓ_b; and $\sigma \in [e^{-\ell_b}, \sigma_{\rm eff}]$. Boxes are bisected in ℓ as in the other variables (the widest of $\Delta\log\mu$, $0.7\Delta\log r$, $0.7\Delta\log\beta$ and $2\Delta\ell$). The μ range is complete: the lower tail with the corners above, $[e^{-30}, 10^{-4}]$, the 16 boxes of v1.0 and the upper tail. *Skip rule (K5).* A box with $\beta_a \geq 10^4 e^{2(\ell'_b-\ell_a)}$, where ℓ'_b is the box's own upper end of ℓ and ℓ_a the column's lower end, contains only states with $b \geq 10^4\sigma^2_{\rm top}$. These belong to the v1.0 boxes or to the μ tails, and the box is skipped. *Rounding.* The number of operands per term does not grow ($2\log\sigma - \log b$ becomes $-\log\beta$). $\log\sigma_{\rm eff}$ is computed with three operations and moved outwards by 10^{-9}. The quantities that depend on ℓ are computed from the box's own floating ends with at most six further operations; their error, below $20u$ relative, lies inside the allowance $1000Su$ of every term.

Checks of the new steps [numerical]. Before use, the new inequalities were attacked with random boxes and states.

- The functions used for the tails and the wedge are generated from the v1.0 term table by a fixed list of textual replacements (`cq_make.py`). In the form of the v1.0 boxes they agree with the v1.0 functions bit for bit on 42000 boxes.

- The components (J2)–(J5), the corner claim, the ends of γ and of the window constants, and the skip rule hold on 10^4 to 9×10^4 random points each.

- Accepted boxes contain no negative state, with the target drawn at random: 2.2×10^4 states in the tails of (ii), 7.2×10^3 in those of (i), 1.4×10^4 in the wedge.

- Ten deliberately wrong versions are all broken: the ends of (J2), (J4) or (J5) swapped; the power shift of (J1) forgotten; the corner $G=0$ forgotten; Π^* at the wrong end; γ or the window constants at the wrong end of ℓ; the skip rule at the wrong end; the τ factors with $\beta = 0.6\tau$.

(`cp_checks.py`, `cq_checks.py`.)

Appendix F: The search for a joint brightness bound

The local envelope of Sec. IX A needs a bound of the brightness P_θ that depends on the spread κ. Before trying to prove a candidate we tried to break it, with the search plan written before the computation and with deliberately wrong versions as a check of the search's power [numerical] (`cn_search.py`). All local quantities are computed from exact Fourier sums.

Candidates. The first form, $P_\theta \leq \min(W,\Psi)\min(1,\sqrt{2\kappa})$ with $W = 2\mu_\theta\bar{Q}^2$, broke at $\mu_\theta = 0.015$ (focus on the crest of the fundamental, a narrow packet on top), with ratio 1.58. The form

$$P_\theta \leq \min\bigl(2\mu_\theta\bar{Q}^2,\ \Psi(\mu_\theta,b)\min(1,\sqrt{2\kappa})\bigr) \tag{F1}$$

survived six scans (about 3.6×10^4 states and hill climbing), with largest ratio 0.998.

Families.

- Coherent packets up to the limit of positivity, at the node, crest and trough of the fundamental; foci at the centre, at $\pm\sqrt{\tau}$ and at a distant node; widths 1 to 256; cosine and sine; five values of x, four of μ_0, four of b (30144 evaluations).

- Narrow packets at small x with a flat band (576) and with a Gaussian envelope in n (216, κ down to 10^{-6}).

- Two packets; combs with teeth from $0.1x$ to $100x$; 3000 random bands; hill climbing (6 candidates × 27 starts × 200 steps).

Power of the search. Four deliberately wrong versions were all broken:

- "$\kappa \geq 1/2$" by a factor 2.9×10^5;
- $\sqrt{\kappa}$ replaced by κ, by 23.4;
- the constant 0.3, by 3.6;
- the constant divided by 10, by 2.0.

The second needed the Gaussian-envelope packets. A packet with a flat band has tails $1/s^2$, its κ is set by the tails, and it looks like $P \propto \kappa$; with a Gaussian envelope the ratio grows like $\kappa^{-1/2}$ as $\kappa \to 0$. For $\mu_\theta \geq 2$ and small κ the largest ratio of the surviving form is 0.099, i.e. a true constant near 0.14 in $P_\theta \leq C\Psi\sqrt{\kappa}$. The proved bound Ψ_2 of Sec. IX A has the constant 25–50 there. The loss comes from comparing E_0 at the two foci, $E_0^{(\tau')} \leq \sqrt{\tau/\tau'}E_0$, which enters the exponent.

Picture. A packet narrower than the focus is, over the whole period, darker by its width. Inside the focus its brightness is the height of the focus times its whole energy, so a concentrated packet obeys $P_\theta \propto \sqrt{\kappa}$.

[1] T. Namba, *How fine can a hidden ripple be? A two-sided logarithmic law for the resolution gap of the arrow of time*, v1.5, Zenodo (2026), doi:10.5281/zenodo.23856796.